

# SDE Interview Preparation: Comprehensive DBMS Q&A

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This document compiles the top 50 Database Management System (DBMS) interview questions, structured for Software Development Engineer (SDE) interview preparation. It is divided into three sections:

- **Top 10 Questions:** The absolute core fundamentals.
  - **Top 25 Questions (Questions 11-25):** Core intermediate topics, SQL querying mechanics, and concurrency.
  - **Top 50 Questions (Questions 26-50):** Advanced optimization, system architecture, index internals, and deep system design concepts.
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## Part 1: Top 10 Core DBMS Questions

### Q1: What are the advantages of using a DBMS over a File System?

#### Answer (from given info):

A DBMS is better than a file system because it organizes data efficiently, reduces redundancy, and ensures secure, reliable, and consistent data management. It supports concurrent multi-user access, reliable transaction management (ACID), efficient data retrieval using SQL, and better scalability. In traditional file systems, the absence of indexing makes search slow (full-page scanning required), files have duplicate/redundant data (leading to inconsistency), and there is a lack of concurrency control (one operation can lock the entire page). DBMS also provides data isolation, data independence, security (authentication and encryption), and automatic backup/recovery.

#### Polished answer:

A Database Management System (DBMS) is an integrated software suite designed to define, store, manipulate, and manage data, serving as a secure interface between applications and physical storage. Compared to traditional File Systems, a DBMS solves critical data management challenges:

1. **Redundancy & Inconsistency Mitigation:** File systems store data in isolated files, leading to duplicate information across directories. If one file is updated and another isn't, data becomes inconsistent. DBMS uses relational schemas and normalization to centralize data, enforcing single-point updates.

2. **Concurrent Multi-User Access:** File systems lack granular concurrency control; multiple users writing to the same file can corrupt it, or one operation may lock an entire file/page. DBMS utilizes sophisticated concurrency control mechanisms (like locking and 2-Phase Locking) to allow thousands of users to read and write simultaneously.
3. **Transactional Integrity (ACID):** A DBMS guarantees that multiple database operations either succeed completely or fail gracefully as a single atomic unit, preserving database state in the face of system crashes.
4. **Declarative Querying & Indexing:** Rather than writing procedural code to scan and filter flat files line-by-line ( $O(N)$  search), a DBMS allows users to write declarative SQL. The engine optimizes execution paths using index structures (such as B+ Trees) for logarithmic  $O(\log N)$  retrieval.
5. **Data Independence:** Physical data independence separates the conceptual schema from the physical storage details. Logical data independence decouples the application programs from the logical schema, allowing structural updates without rewriting application code.

### TL;DR:

A DBMS replaces file systems by centralizing data to eliminate redundancy, enabling safe multi-user concurrency, providing declarative SQL querying with indexing, and guaranteeing transactional safety via ACID properties.

### Keyword/Key mappings:

- DBMS: Database Management System
  - Redundancy: Duplicate data leading to inconsistencies
  - Concurrency: Multiple operations executing simultaneously
  - ACID: Atomicity, Consistency, Isolation, Durability
  - Data Independence: Separation of schema representation from physical storage
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## Q2: What are the ACID properties in DBMS? Explain each.

### Answer (from given info):

ACID stands for Atomicity, Consistency, Isolation, and Durability. These are the key properties that guarantee reliable transaction processing and secure sharing of data among multiple users:

- **Atomicity:** All operations in a transaction are treated as a single unit. If one operation fails, the entire transaction fails, and the database remains unchanged ("all or nothing").
- **Consistency:** Ensures the database starts and ends in a consistent state, with all rules and constraints enforced before and after a transaction.
- **Isolation:** Transactions execute independently; intermediate states of an ongoing transaction are invisible to other transactions.
- **Durability:** Once a transaction is committed, its changes are permanent and will not be lost, even during system failures or restarts.

**Polished answer:**

In transaction processing, a single logical unit of work is called a transaction. The ACID framework guarantees database robustness and accuracy:

1. **Atomicity (All-or-Nothing):** Guarantees that all DML statements within a transaction boundary (between BEGIN and COMMIT) are executed successfully. If any statement fails (due to constraint violations, power loss, or syntax errors), the entire transaction is rolled back using the transaction log (undo phase), leaving the database in its pre-transaction state.
2. **Consistency (State Validity):** Ensures a transaction transitions the database from one valid state to another, maintaining database invariants. All schema constraints (such as primary keys, foreign keys, unique constraints, and check conditions) must hold true at the end of the transaction.
3. **Isolation (Concurrency Control):** Ensures that concurrent execution of transactions results in a system state equivalent to if they were run serially. The intermediate, uncommitted states of a transaction are hidden from other concurrent transactions to prevent data corruption.
4. **Durability (Permanence):** Guarantees that once a transaction commits, its modifications are written to non-volatile storage (e.g., disk). This is achieved via write-ahead logging (WAL), ensuring that committed changes can be reconstructed (redo phase) during crash recovery.

**TL;DR:**

ACID represents the fundamental gold standard of transaction security: Atomicity ensures completeness; Consistency maintains rules; Isolation isolates concurrent operations; Durability guarantees permanent persistence.

**Keyword/Key mappings:**

- Transaction: A single logical unit of database work
  - WAL: Write-Ahead Logging
  - Rollback: Reverting uncommitted changes
  - Commit: Permanently saving modifications
  - Isolation Levels: Settings that control trade-offs between concurrency and isolation
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### **Q3: Explain the Three-Schema Architecture in DBMS and its levels.**

**Answer (from given info):**

The Three-Schema Architecture separates the database into three levels to provide data abstraction and data independence:

- **External Level (View Level):** Defines how different users view the database. Each user sees only the relevant data while hiding the details of the table schema and physical storage.
- **Conceptual Level (Logical Level):** Describes the overall logical structure of the database, including tables, relationships, and constraints. This is the level where developers and system administrators work.

- **Internal Level (Physical Level):** The lowest level, managed by the DBMS, which describes how data is physically stored on disk, including file organization, indexing, and storage paths.

**Polished answer:**

The Three-Schema Architecture (also known as the ANSI-SPARC architecture) is a framework for database design aimed at separating the user applications from the physical database. It implements three distinct levels of abstraction:

1. **External Schema / View Level:** The topmost layer, consisting of several user views. It provides tailored subsets of the database to specific users or applications, hiding unrelated data and schema structures for security and simplicity.
  2. **Conceptual Schema / Logical Level:** The intermediate level that defines the global logical design of the entire database. It specifies what data is stored, the attributes, relationships (such as foreign keys), constraints, and security rules. It is independent of how data is physically stored and how it is viewed by individual users.
  3. **Internal Schema / Physical Level:** The bottommost layer that describes the physical storage structures and access paths. It defines how data blocks are laid out on disk, file organization methods, and physical index structures (e.g., B+ Trees).
- **Physical Data Independence:** The ability to modify the physical schema (e.g., changing disk storage or adding indexes) without changing the conceptual schema.
  - **Logical Data Independence:** The ability to modify the conceptual schema (e.g., adding a table column) without altering the external schemas or application programs.

**TL;DR:**

The Three-Schema Architecture splits a database into External (views), Conceptual (logical design), and Internal (physical storage) levels to maximize data abstraction and maintain both physical and logical data independence.

**Keyword/Key mappings:**

- View Level: User-specific, customized interfaces
- Logical Level: Entity definitions and schema relationships
- Physical Level: Disk block storage and indexes
- Data Abstraction: Hiding implementation details from the user

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## Q4: What is the difference between DBMS and RDBMS?

**Answer (from given info):**

- **DBMS:** Stores data as files or in a non-relational form. It does not support relationships between data, does not enforce integrity constraints, and is suited for smaller/simpler applications. Examples include Microsoft Access, XML Databases, and File Systems.

- **RDBMS:** Relational DBMS stores data in tables (rows and columns). It supports relationships using foreign keys and enforces database integrity constraints (primary and foreign keys). Examples include MySQL, Oracle, SQL Server, and PostgreSQL.

**Polished answer:**

While both manage databases, a Relational Database Management System (RDBMS) is a highly specialized evolution of a basic Database Management System (DBMS):

1. **Data Model:** A basic DBMS stores data in flat files, hierarchical trees, or graph-like network structures. An RDBMS stores data in tabular relations (tables) consisting of rows (tuples) and columns (attributes), mathematically grounded in relational algebra.
2. **Relationships:** A DBMS lacks native capabilities to link records across files or structures systematically. An RDBMS establishes logical relationships between tables using keys (Primary Keys and Foreign Keys).
3. **Integrity Constraints:** Basic DBMS systems do not enforce relational integrity natively, leaving the logic to application code. RDBMS engines strictly enforce entity integrity (e.g., non-null primary keys) and referential integrity (foreign key constraints) at the database level.
4. **ACID Compliance & Concurrency:** RDBMS engines are fundamentally designed to support complex, highly concurrent multi-user ACID transactions. Basic DBMS systems typically have simple locking models (or none) and lack robust transaction managers.
5. **Data Access:** DBMS systems retrieve data programmatically or through basic file operations. RDBMS utilizes SQL (Structured Query Language), a declarative language that allows complex joins and set-based operations.

**TL;DR:**

An RDBMS is a modern, relational subset of DBMS that organizes data into structured tables, natively enforces referential integrity via primary/foreign keys, supports complex SQL joins, and guarantees transactional ACID compliance.

**Keyword/Key mappings:**

- Tabular relations: Structured tables in RDBMS
- Referential integrity: Keeping links between tables consistent
- Declarative querying: Expressing "what" data to retrieve, not "how"

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## Q5: Explain the different types of keys in a database.

**Answer (from given info):**

There are several types of keys used to identify records and maintain relationships:

- **Candidate Key:** A set of one or more attributes that can uniquely identify a row in a table. It has no redundant attributes (minimal superkey).

- **Super Key:** A set of attributes that uniquely identifies a tuple. It is a superset of candidate keys and may contain unnecessary attributes.
- **Primary Key:** A unique identifier chosen from candidate keys to uniquely identify every row. It cannot contain NULL values.
- **Unique Key:** Similar to a primary key but allows NULL values. A table can have multiple unique keys.
- **Alternate Key:** Candidate keys that were not chosen as the primary key.
- **Foreign Key:** An attribute in a table that references the primary key of another table, creating a relationship and maintaining referential integrity.
- **Composite Key:** A combination of two or more columns grouped together to uniquely identify each tuple in a table.

### Polished answer:

Keys are fundamental constraints used to establish identity, maintain uniqueness, and enforce referential integrity across relational tables:

1. **Super Key:** Any set of one or more attributes whose values collectively identify a tuple uniquely within a relation. It is the broadest category of key and may contain redundant attributes.
2. **Candidate Key:** A minimal Super Key. It is a set of attributes that uniquely identifies a row, but if any attribute is removed, it loses its uniqueness property. It contains no redundant attributes.
3. **Primary Key:** The specific Candidate Key chosen by the database designer to uniquely identify rows in a table. It is enforced with **NOT NULL** and **UNIQUE** constraints. A table can have exactly one Primary Key.
4. **Alternate Key:** Any Candidate Key that was not selected as the Primary Key.
5. **Unique Key:** A constraint that ensures all non-null values in a column are distinct. Unlike the Primary Key, it permits **NULL** values (or a single **NULL** depending on the SQL standard/implementation) and multiple Unique Keys can exist per table.
6. **Foreign Key:** A logical link consisting of one or more columns in a child table that references a Primary Key or Unique Key in a parent table. It enforces referential integrity.
7. **Composite Key:** A key that consists of two or more columns. It is used when a single attribute is insufficient to guarantee uniqueness (e.g., in a join table representing a many-to-many relationship).

### TL;DR:

A Super Key is any unique identifier; a Candidate Key is a minimal Super Key; the Primary Key is the chosen unique identifier; Alternate Keys are the unchosen ones; Unique Keys allow uniqueness with NULLs; Foreign Keys link tables; and Composite Keys combine columns.

### Keyword/Key mappings:

- Minimal Super Key: Candidate Key
- Referential Constraint: Foreign Key
- Uniqueness with NULL: Unique Key

- Primary Key: Singular, non-nullable unique identifier
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## Q6: What is a JOIN in SQL? Name and explain different types of joins.

### Answer (from given info):

A JOIN is an operation that combines columns from two or more tables based on a related column between them.

Types of joins:

- **INNER JOIN:** Returns only the rows where there is a match in both tables.
- **LEFT JOIN (LEFT OUTER JOIN):** Returns all rows from the left table and matching rows from the right. If no match is found, NULL values are returned for the right table's columns.
- **RIGHT JOIN (RIGHT OUTER JOIN):** Returns all rows from the right table and matching rows from the left. If no match is found, NULL values are returned for the left table's columns.
- **FULL OUTER JOIN:** Returns all rows when there is a match in either the left or the right table, with NULLs filled in where there are no matches.
- **CROSS JOIN:** Returns the Cartesian product of two tables (every combination of rows from both tables).
- **SELF JOIN:** A join where a table is joined with itself, using aliases to compare rows within the same table.

### Polished answer:

In relational databases, data is normalized and spread across multiple tables. A JOIN is a relational algebra operation that dynamically correlates records across tables using common attributes:

1. **INNER JOIN:** Filters out all non-matching rows. It evaluates the join predicate and returns only the rows where the predicate is satisfied in both relations.
2. **LEFT OUTER JOIN:** Preserves all records from the left relation. For each row in the left table, it attempts to find matching rows in the right table. If a match is found, it populates the columns; otherwise, it outputs **NULL** for the right table's attributes.
3. **RIGHT OUTER JOIN:** The mirror image of the LEFT JOIN. It preserves all records from the right relation and fills left-side columns with **NULL** when there is no match.
4. **FULL OUTER JOIN:** Combines the behavior of LEFT and RIGHT joins. It preserves all records from both tables, populating columns with actual values where matches exist and using **NULL** for missing associations.
5. **CROSS JOIN:** Produces a Cartesian product. If Table A has M rows and Table B has N rows, the result has M \* N rows. It does not use a join predicate.
6. **SELF JOIN:** Syntactically, it is any standard join (INNER or OUTER) where a table is joined to itself. It requires the use of table aliases (e.g., **Employee E1 JOIN Employee E2**) to resolve column references, commonly used for parsing hierarchical self-referential relationships (like finding an employee's manager).

## TL;DR:

SQL Joins combine columns across tables: INNER JOIN returns matches only; LEFT/RIGHT JOIN preserve the left/right table respectively; FULL JOIN preserves all data with NULL placeholders; CROSS JOIN returns a Cartesian product; and SELF JOIN joins a table to itself.

## Keyword/Key mappings:

- Join Predicate: The comparison condition (e.g., ON A.id = B.id)
  - Cartesian Product: Every row combined with every other row (CROSS JOIN)
  - Table Aliases: Temporary names used to reference a table twice in a SELF JOIN
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## Q7: What is meant by normalization and denormalization? How do they differ?

### Answer (from given info):

- **Normalization:** The process of organizing data in a database to reduce redundancy and improve data integrity. It divides large tables into smaller related tables and establishes relationships between them, avoiding anomalies during insertion, update, and deletion.
- **Denormalization:** The process of combining normalized tables back into a single table to introduce controlled redundancy, with the goal of improving query read performance by avoiding expensive join operations.

### Polished answer:

Normalization and Denormalization represent a classic trade-off in database design between write-safety/storage efficiency and read performance:

- **Normalization (Optimizing for Writes and Integrity):** Normalization systematically applies formal design rules (Normal Forms) to decompose large tables. It aims to eliminate redundant data (minimizing storage overhead) and prevent update, insertion, and deletion anomalies. By structuring data atomically, it guarantees that any fact is stored in exactly one place, making updates fast and consistent.
- **Denormalization (Optimizing for Reads):** Denormalization is an optimization technique applied to production systems experiencing heavy read traffic. By pre-joining tables and storing duplicate attributes, it eliminates expensive runtime JOIN operations. While it improves read speed, it introduces data redundancy, which increases storage requirements and places the burden of keeping duplicate data consistent on the application layer.

## TL;DR:

Normalization structures databases into smaller tables to prevent redundancy and write anomalies. Denormalization purposefully duplicates data to optimize read performance and avoid complex JOIN operations in read-heavy applications.

## Keyword/Key mappings:

- Write Anomalies: Inconsistencies during insert, update, or delete operations
  - Redundancy: Duplicate storage of identical facts
  - Relational Decomposition: Splitting tables during normalization
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## Q8: What is the difference between DELETE, TRUNCATE, and DROP in SQL?

### Answer (from given info):

- **DELETE:** A DML command used to delete specific rows from a table based on a WHERE condition. It locks rows, logs each row deletion in the transaction log, is slower, and can be rolled back inside a transaction (depending on database configurations). It fires associated database triggers.
- **TRUNCATE:** A DDL command that removes all rows from a table. It does not maintain individual row logs, cannot be rolled back easily (though supported in some databases like SQL Server/PostgreSQL), does not fire triggers, and is much faster than DELETE because it deallocates the table's data blocks.
- **DROP:** A DDL command that removes the entire table, including its data, schema structure, indexes, and triggers from the database. Once dropped, it cannot be recovered easily.

### Polished answer:

These three commands differ fundamentally in their SQL category (DDL vs. DML), performance, logging overhead, and systemic impact:

#### 1. DELETE (Data Manipulation Language - DML):

- **Operation:** Evaluates row-by-row and deletes records that match a **WHERE** clause.
- **Logging & Transaction:** Every deleted row is logged in the transaction log (Write-Ahead Log), allowing a complete rollback of the transaction.
- **Locks & Triggers:** Acquires row-level locks and fires any configured **DELETE** triggers.
- **Performance:** Slower because of the high transactional logging overhead.

#### 2. TRUNCATE (Data Definition Language - DDL):

- **Operation:** Removes all rows from a table. Instead of scanning and deleting rows, it deallocates the data pages associated with the table.
- **Logging & Transaction:** It only logs the page deallocations, making it extremely lightweight. In some engines (MySQL), it commits implicitly, while in others (PostgreSQL, SQL Server), it can be rolled back if executed within an active transaction block.
- **Locks & Triggers:** Acquires a table-level lock, bypassing row-level locking. It does *not* fire triggers.
- **Space:** Resets auto-increment sequences and releases space back to the database.

#### 3. DROP (Data Definition Language - DDL):

- **Operation:** Destroys the table entirely. It deletes the table's definition, data, constraints, indexes, triggers, and privileges from the database catalog.
- **Rollback:** Cannot be rolled back once executed (unless under transactional DDL support).

#### TL;DR:

**DELETE** is a row-by-row, fully logged DML operation with filtering. **TRUNCATE** is a fast DDL command that empties a table by deallocating pages. **DROP** is a DDL command that permanently deletes the entire table structure and its data.

#### Keyword/Key mappings:

- DML: Data Manipulation Language (operates on rows)
  - DDL: Data Definition Language (operates on schema structures)
  - Page Deallocation: Releasing disk space directly rather than row-by-row clearing
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## Q9: What is an index in DBMS and what are its major types?

#### Answer (from given info):

An index is a data structure (like B-trees, B+ trees, or Hash tables) that improves the speed of data retrieval operations on a database table. It works like a table of contents in a book, allowing the DBMS to quickly find the location of a record without scanning the entire table.

Major types of indexes include:

- **Primary Index:** Created on a sorted data file, usually on the primary key.
- **Clustered Index:** Determines the physical order of data in the table (only one allowed per table).
- **Non-clustered Index:** Creates a separate structure from the table that holds keys and pointers to the actual data rows.
- **Secondary Index:** Created on non-primary columns for faster access.
- **Other types:** B-Tree, B+ Tree, Hash, Bitmap, Full-Text, Composite, Unique, and Covering indexes.

#### Polished answer:

An index is a auxiliary physical data structure designed to accelerate query performance by reducing the disk I/O required to locate specific records. Without an index, the database engine must perform a full-table scan (sequential  $O(N)$  read). With an index, it traverses a structured path (typically  $O(\log N)$ ):

- **B+ Tree Index (Default):** A self-balancing, multi-way search tree. All actual data or row pointers reside in the leaf nodes, which are linked together, enabling fast range scans and point queries.
- **Hash Index:** Uses a hash function to map keys to bucket addresses. It provides  $O(1)$  time complexity for equality searches (e.g., `WHERE id = 5`), but is completely useless for range queries (e.g., `WHERE age > 25`) as it does not preserve sort order.
- **Clustered Index:** The index that defines the physical order of data storage on disk. Since physical data can only be sorted in one way, there can be only one clustered index per table (typically

created on the Primary Key).

- **Non-Clustered Index:** A separate structure that contains the indexed key values and a "row identifier" (RID) or clustering key pointer to the actual data block. You can create multiple non-clustered indexes on a table.
- **Covering Index:** A non-clustered index that contains all the columns referenced in a query's `SELECT`, `WHERE`, and `JOIN` clauses. It allows the database engine to return data directly from the index node without performing an expensive lookup on the base table.

### TL;DR:

An index is an auxiliary data structure (typically a B+ Tree) that speeds up data retrieval. Clustered indexes dictate physical row storage, while non-clustered indexes create separate pointer maps to rows. Covering indexes return data entirely from the index structure.

### Keyword/Key mappings:

- Full Table Scan: Reading all pages from disk sequentially
  - B+ Tree: The industry-standard balanced tree for database indexing
  - Covering Index: An index containing all columns requested by a query
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## Q10: What is a deadlock in DBMS? How can it be prevented or resolved?

### Answer (from given info):

A deadlock occurs when two or more transactions are blocked because each transaction is waiting for a resource that is locked by the other transaction. This results in a situation where none of the transactions can proceed.

- **Prevention Techniques:**
  - **Lock Ordering:** Ensuring that all transactions acquire locks in the same predefined order.
  - **Timeouts:** Automatically rolling back transactions that have been waiting too long for a lock.
- **Resolution Techniques:**
  - **Deadlock Detection:** Periodically checking for cycles using a wait-for graph. If a cycle is detected, the DBMS aborts and rolls back one of the transactions (the "victim"), freeing its locked resources so other transactions can continue.

### Polished answer:

A deadlock is an impasse that arises in highly concurrent environments when two or more transactions hold exclusive locks on resources while requesting locks on resources held by each other, creating a circular wait dependency.

- **Deadlock Prevention (Proactive):**
  - **Strict Lock Ordering:** Standardizing code execution so that all transactions request resources in a globally defined, consistent order (e.g., Table A first, then Table B). This breaks the circular wait condition.

- **Conservative 2-Phase Locking:** Requiring transactions to acquire all necessary locks before starting execution, though this drastically reduces concurrency.
- **Wait-Die / Wound-Wait Schemes:** Utilizing transaction timestamps to decide which transaction waits or dies when a conflict occurs.
- **Deadlock Detection & Resolution (Reactive):**
  - **Wait-For Graph (WFG):** The DBMS maintains a directed graph where nodes represent active transactions and edges represent lock dependencies (Transaction A → Transaction B means A is waiting for B to release a lock).
  - **Detection Cycle:** A background thread periodically runs a cycle-detection algorithm (e.g., Depth-First Search) on the WFG.
  - **Resolution (Victim Selection):** When a cycle is detected, the transaction manager selects a "victim" based on specific heuristics (e.g., transaction age, number of modifications, or cost of rollback). The selected transaction is aborted and rolled back, releasing its locks and allowing the remaining transactions to proceed.

### TL;DR:

A deadlock is a circular wait condition where transactions block each other indefinitely. It is prevented by enforcing strict lock ordering, and resolved by the DBMS running cycle-detection on a Wait-For Graph and rolling back a chosen "victim" transaction.

### Keyword/Key mappings:

- Circular Wait: The core condition for deadlock
- Wait-For Graph (WFG): Directed graph representing lock dependencies
- Victim Selection: Choosing which transaction to abort to break a deadlock

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## Part 2: Top 25 DBMS Questions (Core Intermediate Topics)

### Q11: Explain transaction isolation levels and the concurrency problems they prevent.

#### Answer (from given info):

When multiple transactions execute concurrently, they can cause three major concurrency anomalies:

- **Dirty Read:** A transaction reads uncommitted data modified by another transaction. If that transaction rolls back, the read data is invalid.
- **Non-Repeatable Read:** A transaction reads the same row twice, but gets different values because another transaction updated and committed the row in between.

- **Phantom Read:** A transaction executes a range query twice, but gets a different set of rows because another transaction inserted or deleted rows in that range in between.

To resolve these, SQL databases define four isolation levels:

1. **Read Uncommitted:** Lowest isolation. Fast but permits dirty, non-repeatable, and phantom reads.
2. **Read Committed:** Prevents dirty reads. Still allows non-repeatable and phantom reads. Default in PostgreSQL.
3. **Repeatable Read:** Prevents dirty and non-repeatable reads. Still allows phantom reads (prevented in some engines like MySQL). Default in MySQL.
4. **Serializable:** Highest isolation. Prevents all anomalies by executing transactions with strict concurrency restrictions (slowest).

### Polished answer:

Transaction isolation levels define the degree to which data modifications made by one transaction are visible to other concurrent transactions. They balance data consistency against system performance:

#### 1. Read Uncommitted:

- **Mechanism:** Read operations do not acquire shared locks, allowing transactions to read uncommitted "dirty" modifications.
- **Anomaly:** Permitted Dirty Reads, Non-Repeatable Reads, and Phantom Reads.

#### 2. Read Committed:

- **Mechanism:** Shared locks are released as soon as the read operation completes (instead of holding them until the end of the transaction). It ensures reads only access committed data.
- **Anomaly:** Prevents Dirty Reads; permits Non-Repeatable Reads and Phantom Reads.

#### 3. Repeatable Read:

- **Mechanism:** Shared locks acquired for reading are held until the transaction finishes.
- **Anomaly:** Prevents Dirty and Non-Repeatable Reads; permits Phantom Reads (though engines like InnoDB in MySQL use Multi-Version Concurrency Control (MVCC) and Next-Key locking to prevent phantoms at this level).

#### 4. Serializable:

- **Mechanism:** Implements range locking or locking of entire tables (or equivalent serializability checks in optimistic models) to completely isolate transactions.
- **Anomaly:** Prevents all anomalies (Dirty, Non-Repeatable, and Phantom Reads) but drastically limits transaction throughput.

### TL;DR:

SQL transaction isolation levels (Read Uncommitted, Read Committed, Repeatable Read, and Serializable) prevent concurrency anomalies (Dirty, Non-Repeatable, and Phantom Reads) by varying lock durations and concurrency control strategies.

### Keyword/Key mappings:

- MVCC: Multi-Version Concurrency Control
- Concurrency Anomalies: Data inconsistencies due to concurrent execution

- Next-Key Lock: Lock mechanism to prevent phantom reads
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## Q12: What is the difference between WHERE and HAVING clauses in SQL?

### Answer (from given info):

- **WHERE:** Used to filter rows *before* any grouping or aggregation takes place. It operates on individual records and cannot contain aggregate functions (like SUM, COUNT, AVG).
  - **HAVING:** Applied *after* the GROUP BY clause to filter aggregated summary rows. It is meant for filtering grouped data based on aggregate functions.
- Using HAVING instead of WHERE for simple filtering makes queries slower because unnecessary rows are grouped first and then filtered out later.
- The order of SQL query execution is: FROM/JOIN -> WHERE -> GROUP BY -> HAVING -> SELECT -> ORDER BY -> LIMIT .

### Polished answer:

The distinction between WHERE and HAVING resides entirely in their position in the logical query processing pipeline:

1. **Evaluation Phase:** WHERE is evaluated early in the pipeline, immediately after the tables are loaded via FROM / JOIN . It filters individual raw rows out of the dataset before any grouping occurs. HAVING is evaluated much later, after the GROUP BY clause has partitioned the remaining rows into aggregate buckets.
2. **Aggregations:** Because WHERE runs before grouping, it cannot evaluate aggregate functions (e.g., WHERE COUNT(order\_id) > 5 will throw a syntax error). Conversely, HAVING is designed specifically to filter on aggregate properties (e.g., HAVING SUM(amount) > 1000 ).
3. **Performance Optimization:** If a filter does not involve aggregate functions, it should *always* be placed in the WHERE clause. Filtering early with WHERE reduces the volume of data that the database engine must sort, partition, and aggregate in memory. Placing a row-level filter in the HAVING clause forces the database to aggregate unnecessary rows, severely degrading query performance.

### TL;DR:

WHERE filters individual rows *before* grouping occurs and cannot contain aggregates. HAVING filters aggregated groups *after* GROUP BY has executed. For optimal performance, always filter row-level conditions using WHERE .

### Keyword/Key mappings:

- Logical Query Processing: The sequence of step-by-step evaluations SQL uses to execute a query

- Row-level filter: Filtering applied by WHERE
  - Group-level filter: Filtering applied by HAVING
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## Q13: What is a view in DBMS? How does it differ from a table and when should you avoid it?

### Answer (from given info):

A View is a virtual table defined by a SQL query that accesses data from one or more base tables. It does not store data physically on disk (unlike a real table), but dynamically runs its underlying query whenever the view is accessed.

- **Benefits:** Simplifies complex queries, provides a security layer by hiding specific columns or rows, and maintains a consistent interface for users.
- **When to Avoid:** Avoid using views for heavy calculations or frequently accessed large datasets. Views do not improve read performance because their underlying query must execute every time the view is queried, which can hide inefficient SQL and make debugging difficult. In some cases, direct inserts or updates through views are restricted.

### Polished answer:

A view is a logical abstraction layer over physical tables. It is defined as a stored SQL query in the database catalog that acts as a read-only or limited-writable virtual relation:

- **Conceptual Differences:** A base table has physical data storage (data blocks, rows, pages, and clustered indexes). A standard view is purely metadata—it stores only the SQL definition. When queried, the engine merges the view's query with the outer query and optimizes them together.
- **When to Avoid Views:**
  1. **Heavy Computations:** If a view contains complex calculations, subqueries, or expensive joins, querying it repeatedly causes high CPU and memory usage because the calculations are re-evaluated on every access.
  2. **Layered Views on Views:** Creating "nested views" (views querying other views) can result in highly inefficient and unoptimized execution plans, hiding nested table scans that make query optimization and debugging extremely difficult.
  3. **Write-Heavy Workflows:** DML operations (INSERT, UPDATE, DELETE) through views are highly restricted. If a view joins multiple tables, contains aggregations ( `GROUP BY` ), or lacks key attributes from the underlying base tables, it is non-updatable.

### TL;DR:

A view is a virtual table representing a stored SQL query. It simplifies complex joins and improves security, but it should be avoided for computationally intensive tasks or nested architectures because it re-runs its query on every execution, adding performance overhead.

### Keyword/Key mappings:

- Base Table: The physical table containing raw data
  - Dynamic Query Merging: The database optimizer merging view SQL with the main query
  - Nested Views: Defining views on top of other views (anti-pattern)
- 

## Q14: Explain the difference between Clustered and Non-Clustered Indexes.

### Answer (from given info):

- **Clustered Index:** Organizes and physically sorts the actual data rows of the table based on the indexed column values. There can only be one clustered index per table because data rows can only be physically sorted in one way on disk.
- **Non-Clustered Index:** Creates a separate structure from the table that holds the indexed key columns and pointers (row identifiers) to the actual physical data rows. Multiple non-clustered indexes can be created on a table.

### Polished answer:

Clustered and non-clustered indexes are physical structures used by the storage engine to organize and access data:

#### 1. Physical Storage Layout:

- In a **Clustered Index**, the leaf nodes of the B+ Tree contain the *actual data rows* of the table. The index *is* the table.
- In a **Non-Clustered Index**, the leaf nodes of the B+ Tree contain the *indexed key values* along with a locator. In a heap table, this locator is a physical Row ID (RID). In a table with a clustered index, the locator is the corresponding Clustered Index Key.

2. **Cardinality Limitation:** A table can have exactly **one** Clustered Index because data cannot be physically sorted in more than one order on disk. Conversely, a table can have **hundreds** of Non-Clustered Indexes because they are stored as separate auxiliary structures.

#### 3. Lookup Mechanics:

- Searching via a clustered index takes  $O(\log N)$  operations, leading directly to the data.
- Searching via a non-clustered index requires an initial  $O(\log N)$  search on the index tree, followed by an additional lookup (Key Lookup or RID Lookup) to retrieve non-indexed columns from the base table. This additional step can cause high random I/O overhead.

### TL;DR:

A clustered index determines the physical sort order of the table's rows and stores the actual data in its leaf nodes (limit: 1 per table). A non-clustered index is a separate structure containing keys and pointers to the physical data rows (limit: multiple).

### Keyword/Key mappings:

- Heap Table: A table without a clustered index, where rows are stored unordered
  - RID Lookup: Accessing data using a physical disk address pointer
  - Key Lookup: Accessing data using the clustered index key found in a non-clustered index
- 

## Q15: What is a subquery in SQL? What is a correlated subquery, and when is it slower than a JOIN?

### Answer (from given info):

A subquery is a query embedded within another query (inside SELECT, INSERT, UPDATE, or DELETE statements).

- **Correlated Subquery:** A subquery that depends on values from the outer query. It executes repeatedly—once for every single row returned by the outer query—making it generally slower than a normal subquery or a JOIN.

A correlated subquery is slower than a JOIN on large datasets because a JOIN combines tables and executes efficiently in a set-based manner, whereas a correlated subquery acts like a nested loop (  $O(N * M)$  ), evaluating the subquery for every outer row.

### Polished answer:

An SQL subquery (or inner query) is a nested SELECT statement used to return data for use in the primary (outer) query:

- **Non-Correlated Subquery:** Evaluates independently of the outer query. The database engine executes the subquery exactly once, caches its result, and passes it to the outer query.
- **Correlated Subquery:** Refers to columns in the outer query (e.g., `WHERE outer.col = inner.col`). Because of this logical coupling, the inner query cannot be evaluated independently; it must run **once for every row** processed by the outer query.
- **Why and When it is Slower than a JOIN:**
  - **Procedural Execution:** A correlated subquery behaves like a nested loop. If the outer table has N rows and the inner table has M rows, the engine may perform N index or table scans on the inner table, leading to  $O(N * M)$  complexity.
  - **JOIN Performance:** A JOIN is processed set-by-set. The database query optimizer can build highly efficient Hash Joins or Merge Joins, reading both tables into memory, matching keys in a single pass, and utilizing parallel processing.
  - **When to refactor:** If you are querying large tables, correlated subqueries cause severe disk I/O bottlenecks and should be refactored into JOINs or CTEs.

### TL;DR:

A subquery is an inner query. A correlated subquery is logically dependent on the outer query, forcing it to execute once per outer row ( $O(N * M)$  loop). It is slower than a JOIN on large tables because JOINs allow set-based optimizations like Hash or Merge Joins.

## Keyword/Key mappings:

- Set-based processing: Querying data collections in blocks (efficient)
  - Nested Loop: Procedural execution of correlated subqueries (inefficient)
  - Hash Join: High-performance join algorithm utilizing in-memory hash tables
- 

## Q16: What are triggers in DBMS? When are they used and what are their drawbacks?

### Answer (from given info):

A trigger is a special kind of stored procedure that automatically executes (fires) in response to specific events on a table, such as INSERT, UPDATE, or DELETE. They cannot be called manually.

- **Types of Triggers:**
  - **BEFORE trigger:** Runs before the database operation is performed (often used for data validation or modification).
  - **AFTER trigger:** Runs after the operation is completed (commonly used for audit logging or updating related data).
- **Drawbacks:** Triggers can make the database harder to debug because they run automatically behind the scenes, making the execution flow less visible. Chaining multiple triggers can cause unexpected behaviors or performance overhead.

### Polished answer:

Triggers are event-driven database scripts compiled and executed inside the database engine when specific DML modifications occur on target tables:

- **Use Cases:**
  1. **Enforcing Complex Business Rules:** Validating state transitions that cannot be easily expressed with static constraints (like **CHECK** constraints).
  2. **Automated Audit Logging:** Capturing historical changes by writing the previous state ( **OLD** pseudo-record) and the new state ( **NEW** pseudo-record) of a row into an audit table.
  3. **Maintaining Derived Data:** Automatically updating redundant or summary tables (e.g., modifying a user's total balance when a transaction is inserted).
- **Drawbacks:**
  1. **Hidden Side Effects:** Because triggers run implicitly, application developers may be unaware of additional database writes. This "spooky action at a distance" makes tracing errors and debugging highly complex.
  2. **Cascading / Infinite Loops:** An **AFTER UPDATE** trigger on Table A might update Table B, which has a trigger updating Table A. If not configured carefully, this can cause cascading locks, transaction timeouts, or infinite loops.

3. **Performance Overhead:** Triggers execute within the transaction context of the initiating DML statement. If a trigger does heavy processing, it extends lock durations on the target rows, causing concurrency bottlenecks.

#### TL;DR:

Triggers are event-driven scripts that automatically execute on INSERT, UPDATE, or DELETE operations. While excellent for auditing and complex validation, they add hidden performance overhead, complicate debugging, and can cause cascading locks.

#### Keyword/Key mappings:

- Pseudo-records: **NEW** and **OLD** records representing row states in a trigger
  - Audit Logging: Tracking historical changes in data
  - Spooky action at a distance: Implicit execution that is invisible to the application code
- 

## Q17: What are stored procedures and functions? Explain their differences.

#### Answer (from given info):

Stored procedures and functions are precompiled sets of SQL statements stored in the database to reuse logic, reduce network traffic, and improve security.

Differences:

- **Stored Procedure:** Executed explicitly using CALL or EXEC. It is used to perform database operations (INSERT, UPDATE, DELETE). It accepts parameters as IN, OUT, or INOUT, and it is *not* required to return a value. It cannot be used directly inside standard SQL queries (like SELECT statements).
- **Function:** Mainly used to compute and return a single value. It must return a value. It can accept only input parameters (IN) and can be used directly inside SQL statements (e.g., **SELECT GetEmployeeSalary(id) FROM Employee**).

#### Polished answer:

Stored procedures and stored functions are procedural code modules compiled and executed directly within the database engine, reducing client-server roundtrips:

##### 1. Return Values:

- A **Stored Function** must return exactly one value (or a table in table-valued functions) using the **RETURN** keyword.
- A **Stored Procedure** does not require a return value. It can return data using **OUT** or **INOUT** parameters or by returning a dynamic result set (a cursor).

##### 2. SQL Integration:

- Functions can be used inside SQL expressions (e.g., **SELECT** , **WHERE** , **HAVING** ).

- Procedures must be executed using explicit statements like `CALL` or `EXECUTE` and cannot be embedded in standard SQL queries.

### 3. Transaction Management:

- Stored Procedures can manage transactions (they can execute `COMMIT` and `ROLLBACK` commands).
- Functions generally cannot execute transaction control statements because they run inside queries, which must preserve read consistency.

4. **Purpose:** Use functions for deterministic computations (e.g., calculating tax). Use procedures to encapsulate business operations (e.g., processing an e-commerce order).

### TL;DR:

Stored Procedures are designed to perform complex business operations, can manage transactions, and are invoked using `CALL`. Stored Functions are designed to compute and return a single value, and they can be called directly within SQL queries.

### Keyword/Key mappings:

- Deterministic Function: A function that always returns the same output for a given input
  - IN/OUT/INOUT: Parameter modes in stored procedures
  - Precompiled Execution: Pre-analyzing SQL code to execute it faster
- 

## Q18: What is a database lock? Explain Shared Lock (S) vs. Exclusive Lock (X).

### Answer (from given info):

A database lock is a mechanism that protects a shared piece of data from getting updated by multiple database users at the same time to ensure data consistency during concurrent transactions.

- **Shared Lock (S Lock):** Allows multiple transactions to read a data item concurrently, but prevents any transaction from modifying the data until all shared locks are released.
- **Exclusive Lock (X Lock):** Prevents any other transaction from reading or modifying the locked resource. It is acquired by a transaction about to perform a write operation (INSERT, UPDATE, DELETE).

### Polished answer:

Database locks are concurrency control primitives used by the transaction manager to implement isolation levels and prevent conflicts:

- **Shared Lock (S-Lock / Read Lock):** Multiple transactions can hold a Shared Lock on the same database resource (row, page, or table) simultaneously. It guarantees that the resource will not be modified while those transactions are reading it. While an S-lock is active, other transactions can request and obtain S-locks, but any request for an Exclusive Lock (X-lock) is blocked.

- **Exclusive Lock (X-Lock / Write Lock):** Only one transaction can hold an Exclusive Lock on a resource. It is required for any data modification (write, update, delete) operation. When a transaction holds an X-lock, all other transactions are blocked from acquiring *any* lock (Shared or Exclusive) on that resource.
- **Lock Compatibility Matrix:**
  - S-Lock with S-Lock: **Compatible** (allows concurrent reads).
  - S-Lock with X-Lock: **Conflict** (readers block writers, and vice-versa).
  - X-Lock with X-Lock: **Conflict** (writers block writers).

### TL;DR:

A Shared Lock (S-lock) allows concurrent reads but blocks writes. An Exclusive Lock (X-lock) grants sole access to a resource for writing, blocking all other readers and writers to prevent dirty reads and data corruption.

### Keyword/Key mappings:

- Concurrency Control: Managing simultaneous transaction executions without conflicts
- Lock Compatibility: The rules governing whether multiple locks can be held on a resource
- Write Lock: Exclusive lock (X-lock)

## Q19: What is referential integrity in DBMS?

### Answer (from given info):

Referential Integrity ensures that relationships between tables are maintained correctly and consistently. It requires that a foreign key in one table must match a valid primary key (or unique key) in another table, or be NULL. This prevents orphaned records and ensures data consistency across related tables. For example, if a row in the parent table is deleted, referential integrity defines what happens to the child rows (using CASCADE, SET NULL, or RESTRICT).

### Polished answer:

Referential Integrity is a relational database design constraint that guarantees the logical consistency of links between associated tables. It is enforced using the **FOREIGN KEY** constraint:

1. **The Core Rule:** If Table B has a foreign key referencing the primary key of Table A, every non-null foreign key value in Table B must exist as a primary key value in Table A. This ensures that no record points to a non-existent entity (preventing "orphaned" records).
2. **Referential Actions (Cascades):** When a referenced record in Table A is updated or deleted, referential integrity rules dictate how the change propagates to the child records in Table B:
  - **ON DELETE RESTRICT / NO ACTION** : Blocks the deletion of the parent row if child rows exist.
  - **ON DELETE CASCADE** : Automatically deletes all child rows in Table B when the parent row in Table A is deleted.

- **ON DELETE SET NULL** : Sets the foreign key column in Table B to **NULL** when the parent row is deleted.

### TL;DR:

Referential Integrity ensures that relations between tables remain consistent. It dictates that every foreign key must reference a valid primary key in a parent table (or be NULL), preventing orphaned records using actions like CASCADE or RESTRICT.

### Keyword/Key mappings:

- Orphaned Record: A child row whose foreign key points to a non-existent parent row
  - Cascade: Propagating changes automatically from a parent to a child table
  - Referential Constraint: Foreign Key
- 

## Q20: Explain the difference between UNION and UNION ALL from a performance viewpoint.

### Answer (from given info):

- **UNION**: Combines the results of two or more queries and removes duplicate rows from the final output. To do this, the database engine must perform a sort and compare operation on the rows, which consumes extra CPU and memory.
- **UNION ALL**: Combines the results of two or more queries but does *not* check for or remove duplicates. It simply appends the datasets together, making it faster and much more resource-efficient.

If you know the datasets do not overlap (or if duplicates are acceptable), **UNION ALL** should always be preferred for better performance.

### Polished answer:

The difference between **UNION** and **UNION ALL** represents a direct trade-off between output uniqueness and execution performance:

#### 1. Execution Mechanics:

- **UNION**: The database engine retrieves records from both queries, appends them, and then performs a **Duplicate Elimination** step. This typically requires sorting the combined dataset in memory (or on disk if the dataset is large) and applying a unique filter, or building a temporary hash table. This sort/hash operation runs in  $O(N \log N)$  time and can cause severe memory paging.
- **UNION ALL**: The database engine simply streams the results of the first query and then streams the results of the second query directly to the client. It performs no comparisons, sorting, or deduplication. This runs in  $O(N)$  time and requires negligible auxiliary memory.

2. **SDE Best Practice:** Never use `UNION` if you know that the two queries will produce mutually exclusive results (disjoint sets). Use `UNION ALL` to bypass the expensive sorting and deduplication phases.

**TL;DR:**

`UNION` removes duplicates, requiring an expensive in-memory sort or hash deduplication step ( $O(N \log N)$  complexity). `UNION ALL` simply appends the result sets without deduplication ( $O(N)$  complexity), making it significantly faster and less resource-intensive.

**Keyword/Key mappings:**

- Deduplication: Removing duplicate records from a dataset
  - Duplicate Elimination: The sort or hash step executed during UNION
  - Disjoint Sets: Sets that share no common elements
- 

## Q21: What are the different types of relationships in DBMS?

**Answer (from given info):**

There are four major types of relationships amongst tables in a DBMS:

- **One-to-One (1:1):** A record in Table A is associated with exactly one record in Table B. E.g., a person and their passport.
- **One-to-Many (1:M) / Many-to-One (M:1):** A record in Table A can be associated with multiple records in Table B, but a record in Table B is associated with only one record in Table A. E.g., a customer and their orders.
- **Many-to-Many (M:M):** Multiple records in Table A can be associated with multiple records in Table B. E.g., students and courses. This is resolved in databases by using a junction/associative table.
- **Self-Referencing Relationship:** A row in a table is associated with another row in the same table. E.g., an employee table containing a `manager_id` that references the `employee_id` of the same table.

**Polished answer:**

Entity relationships define how tables are logically connected to model real-world business domains:

1. **One-to-One (1:1):** Modelled by placing the Primary Key of Table A as a Unique Foreign Key in Table B. Used to split tables for security, optimization, or subclassing.
2. **One-to-Many (1:N):** The foundation of relational structures. Modelled by placing the Primary Key of the "One" side (parent table) as a standard Foreign Key on the "Many" side (child table).
3. **Many-to-Many (M:N):** Relational databases cannot directly implement many-to-many links between two tables. Instead, an **associative (junction/join) table** is introduced. This bridge table contains foreign keys referencing both tables, usually combined to form a Composite Primary Key.

4. **Self-Referential / Recursive Relationship:** Occurs when an entity has a relationship with itself. Modelled by placing a foreign key within a table that references its own primary key (e.g., `ManagerID` referencing `EmployeeID` in an `Employees` table).

#### TL;DR:

1:1 relations use a unique foreign key; 1:N relations place a foreign key on the "Many" side; M:N relations require a third associative (junction) table to bridge the keys; self-referential relations use a foreign key that references the same table's primary key.

#### Keyword/Key mappings:

- Junction Table: A table used to split an M:N relationship into two 1:N relationships
  - Recursive Foreign Key: A foreign key pointing to its own table
  - Associative Relation: Another name for a junction table
- 

## Q22: What are constraints in SQL? Name and explain the common types.

#### Answer (from given info):

Constraints are rules applied to columns in a database table to limit the type of data inserted, ensuring data accuracy, integrity, and consistency.

The most common SQL constraints are:

- **NOT NULL:** Ensures a column cannot contain NULL values.
- **UNIQUE:** Ensures all values in a column are distinct. It allows NULL values.
- **PRIMARY KEY:** Uniquely identifies each record in a table. It is a combination of NOT NULL and UNIQUE, and only one is allowed per table.
- **FOREIGN KEY:** Links a column in one table to the primary key of another, maintaining referential integrity.
- **CHECK:** Ensures that all values in a column satisfy a specific condition (e.g., `Age >= 18`).
- **DEFAULT:** Assigns a default fallback value to a column if no value is provided during insertion.

#### Polished answer:

Constraints are declarative rules enforced by the relational database engine during write transactions. If a query attempts to insert or modify data in a way that violates a constraint, the engine aborts the statement and rolls back any partial changes:

1. **NOT NULL:** Enforces domain integrity, ensuring that a column must have a value (prevents unknown state).
2. **UNIQUE:** Enforces that no two rows in the table can share the same non-null value in the constrained column(s). The database engine automatically creates an underlying non-clustered index (usually a B-Tree) to verify uniqueness in  $O(\log N)$  time.
3. **PRIMARY KEY:** A physical and logical constraint. It establishes entity integrity by requiring uniqueness and prohibiting NULLs.

4. **FOREIGN KEY:** Enforces referential integrity by validating that child table records have matching parents.
5. **CHECK:** Evaluates a boolean expression for every row insertion or update (e.g., `CHECK (salary > 0)`). If the expression evaluates to false, the write operation is rejected.
6. **DEFAULT:** Sets a fallback value when a column is omitted in an `INSERT` statement, avoiding NULLs and simplifying data entry.

### TL;DR:

Constraints are declarative rules (NOT NULL, UNIQUE, PRIMARY KEY, FOREIGN KEY, CHECK, DEFAULT) enforced by the database engine to maintain data integrity, automatically rolling back any transaction that violates them.

### Keyword/Key mappings:

- Domain Integrity: Validating that data falls within acceptable ranges (NOT NULL, CHECK)
  - Entity Integrity: Ensuring every row is uniquely identifiable (PRIMARY KEY)
  - Declarative Rules: Constraints defined in the schema itself rather than in application logic
- 

## Q23: What is a materialized view and what are its trade-offs?

### Answer (from given info):

A materialized view is a database object that physically stores the results of a query on disk, unlike a standard view which is virtual. It improves query read performance by precomputing and storing the aggregated or joined results.

- **Trade-offs:**
  - **Pros:** Read operations are extremely fast, making it excellent for complex reporting, data warehousing, and analytics queries.
  - **Cons:** The stored data can become stale over time. It requires a refresh mechanism (manual or scheduled) to synchronize with base tables. Refresh operations can be computationally expensive and degrade write performance on the base tables.

### Polished answer:

A materialized view is a physical caching mechanism for SQL query results, commonly used in OLAP (Online Analytical Processing) systems:

- **How it Works:** Rather than parsing and executing a complex SQL query on every request, the database engine executes the query once, writes the result set as a concrete table to disk, and indexes it. Subsequent reads query the cached disk data directly, yielding  $O(1)$  or  $O(\log N)$  read times.
- **Key Trade-Offs:**

1. **Query Performance vs. Data Staleness:** Reads are incredibly fast, but the data is a "snapshot" and becomes stale as the underlying base tables are updated.
2. **Refresh Strategies (Compute Overhead):** To keep data accurate, the view must be refreshed:
  - **Complete Refresh:** Re-runs the entire query from scratch, causing high CPU and disk I/O.
  - **Incremental / Fast Refresh:** Applies only the changes recorded in materialized view logs (WAL-like logs) of the base tables since the last refresh. This is highly complex but efficient.
3. **Write Amplification:** If the view is set to refresh immediately upon base table updates ( `REFRESH ON COMMIT` ), write latency on the base tables increases significantly.

#### TL;DR:

A materialized view caches query results physically on disk. It provides massive speed improvements for complex analytical queries (OLAP) at the cost of data staleness, storage overhead, and compute-heavy refresh cycles.

#### Keyword/Key mappings:

- OLAP: Online Analytical Processing (analytics/reporting)
  - OLTP: Online Transaction Processing (fast, frequent transactions)
  - Refresh on Commit: Updating the materialized view immediately when base data changes
- 

## Q24: Explain the difference between 2-Tier and 3-Tier Client-Server Architecture in a DBMS.

#### Answer (from given info):

- **2-Tier Architecture:** A basic client-server architecture where the client application communicates directly with the database at the server end without any intermediate middleware. E.g., a desktop contact manager created using MS-Access or a local railway reservation system.
- **3-Tier Architecture:** Introduces an intermediate application layer (business logic layer) between the client application and the database server. The client interacts with the application server (via a GUI or web browser), which in turn processes the request and communicates with the database. E.g., modern web applications.

#### Polished answer:

DBMS client-server architectures define the division of labor, security boundaries, and scalability of database applications:

##### 1. 2-Tier Architecture (Client-Server):

- **Structure:** Consists of a Client (Presentation Layer + Business Logic) and a Server (Database Layer). The client application runs SQL queries directly against the database using drivers like JDBC or ODBC.
- **Pros/Cons:** Simple to design for small systems. However, it scales poorly because every client maintains an active database connection. It also poses severe security risks, as database credentials and SQL logic are exposed on the client machine.

## 2. 3-Tier Architecture (Modern Web App Model):

- **Structure:** Divides the system into three logical layers:
  - **Presentation Tier (Client):** The user interface (e.g., mobile app, web browser) that does not interact with the database directly.
  - **Application Tier (Business Logic Middleware):** An application server (e.g., Node.js, Spring Boot) that contains the business logic. It authenticates users, validates requests, and manages a pool of database connections.
  - **Database Tier (Data Store):** The raw database system which only receives queries from the trusted application tier.
- **Pros/Cons:** Highly secure (credentials are hidden), supports load balancing, and scales efficiently because connection pooling is managed at the application server level.

### TL;DR:

A 2-tier architecture features direct communication between the client and the database server. A 3-tier architecture inserts an application server in between to handle business logic, drastically improving security, scalability, and connection pooling.

### Keyword/Key mappings:

- Connection Pooling: Reusing a cache of database connections rather than creating new ones
- Business Logic: The core rules and processing workflows of an application
- Middleware: Software acting as a bridge between an operating system/database and applications

## Q25: Explain SQL vs. NoSQL Databases.

### Answer (from given info):

- **SQL (Relational):** Requires a strict, predefined schema before inserting data. Data is stored in structured tables with rows and columns. It scales vertically (adding more CPU/RAM to a single server) and guarantees ACID properties for transactional consistency. Ideal for highly structured data where relationships are critical (e.g., financial systems). E.g., MySQL, PostgreSQL.
- **NoSQL (Non-Relational):** Features a flexible, dynamic schema. Data can be stored as key-value pairs, documents (JSON), wide columns, or graphs. It scales horizontally (adding more servers to a cluster) and adheres to the CAP theorem. Optimized for large-scale, rapidly changing datasets. E.g., MongoDB, Redis, Cassandra.

### Polished answer:

SQL (relational) and NoSQL (non-relational) databases represent fundamentally different data-modeling and system scaling philosophies:

#### 1. Schema Design & Flexibility:

- **SQL:** Implements a declarative schema. Tables must be defined beforehand, and modifications require schema migrations. Excellent for enforcing data consistency and relationships.
- **NoSQL:** Schema-less or schema-on-read. It natively handles unstructured or semi-structured data, allowing document schemas to change on the fly.

#### 2. Scaling Path:

- **SQL:** Scales **vertically** (scale-up). Scaling involves upgrading hardware on a single node. While sharding (horizontal partitioning) is possible, it is highly complex to implement and maintain.
- **NoSQL:** Scaled **horizontally** (scale-out) by design. Data is partitioned across a distributed cluster of cheap commodity servers.

#### 3. Query Language and JOINS:

- **SQL:** Relies on standardized SQL. It handles complex, multi-table JOINS and nested aggregations efficiently.
- **NoSQL:** Uses API queries (e.g., MongoDB's MQL) or specialized query languages. It does not support native, high-performance JOIN operations; data is typically denormalized (nested) within a single document to optimize read speed.

4. **Consistency Model:** SQL prioritizes **ACID** consistency. NoSQL typically prioritizes **BASE** consistency (Basically Available, Soft state, Eventual consistency) to optimize availability and partition tolerance (guided by the CAP Theorem).

### TL;DR:

SQL databases are structured, relational, ACID-compliant, and scale vertically, making them ideal for complex queries and transaction safety. NoSQL databases are flexible, schema-less, eventually consistent, and scale horizontally, optimized for massive write/read throughput.

### Keyword/Key mappings:

- Vertical Scaling: Upgrading the CPU/RAM of a single machine
- Horizontal Scaling: Adding more server nodes to a distributed cluster
- CAP Theorem: A distributed system can guarantee only two of: Consistency, Availability, and Partition Tolerance

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## Part 3: Top 50 DBMS Questions (Advanced System Mechanics)

## Q26: What are the different Normal Forms in DBMS? Explain 1NF, 2NF, 3NF, and BCNF.

### Answer (from given info):

Normalization organizes data into multiple tables to eliminate redundancy and anomalies:

- **1NF (First Normal Form):** Every column must contain single, atomic (indivisible) values. Multi-valued attributes or repeating groups are forbidden.
- **2NF (Second Normal Form):** Must be in 1NF, and all non-key attributes must be fully functionally dependent on the primary key (no partial dependency where an attribute depends on only a part of a composite primary key).
- **3NF (Third Normal Form):** Must be in 2NF, and there can be no transitive dependency (non-key attributes must not depend on other non-key attributes).
- **BCNF (Boyce-Codd Normal Form):** A stricter version of 3NF (often called 3.5NF). For every functional dependency  $A \rightarrow B$ , the determinant  $A$  must be a super key.

### Polished answer:

Normalization is a theoretical database design framework that decomposes tables to eliminate redundancy-related anomalies (Insert, Update, Delete) based on Functional Dependencies (FDs):

1. **1NF (Atomic Domains):** A relation is in 1NF if and only if the domain of each attribute contains only atomic (indivisible) values, and there are no repeating groups. E.g., a "Subjects" column cannot store a comma-separated list like "Math, Science"; it must be split into separate rows.
2. **2NF (No Partial Dependencies):** A relation is in 2NF if it is in 1NF and every non-prime attribute is fully functionally dependent on the primary key. This is only relevant when the primary key is composite. If a non-prime attribute depends on only *part* of the composite primary key, it violates 2NF and must be decomposed.
3. **3NF (No Transitive Dependencies):** A relation is in 3NF if it is in 2NF and no non-prime attribute is transitively dependent on the primary key. That is, if  $A \rightarrow B$  and  $B \rightarrow C$  (where  $A$  is the primary key and  $B$  is a non-prime attribute), then  $C$  is transitively dependent on  $A$  through  $B$ . This is resolved by splitting  $B$  and  $C$  into a new table.
4. **BCNF (Boyce-Codd / 3.5NF):** A relation is in BCNF if and only if for every non-trivial functional dependency  $A \rightarrow B$ ,  $A$  is a Super Key. BCNF addresses anomalies that 3NF misses when a table has overlapping composite candidate keys.

### TL;DR:

1NF enforces atomic values. 2NF removes partial dependencies on composite keys. 3NF removes transitive dependencies (non-key pointing to non-key). BCNF ensures that for every functional dependency  $A \rightarrow B$ ,  $A$  must be a super key.

### Keyword/Key mappings:

- **Functional Dependency (FD):** A relationship where one attribute uniquely determines another
  - **Non-prime Attribute:** An attribute that is not part of any candidate key
  - **Transitive Dependency:** A non-key column depending on another non-key column
- 

## Q27: What is a CTE and how does it differ from a subquery?

### Answer (from given info):

A CTE (Common Table Expression), written using the `WITH` clause, is a temporary named result set used within a single query execution.

Differences:

- **Readability:** CTEs make complex queries much easier to read and maintain than nested subqueries. They can be referenced multiple times in the same query.
- **Recursive Capabilities:** Recursive CTEs can handle hierarchical data (like org charts or parent-child categories), which standard subqueries cannot do easily.
- **Performance:** Some database engines materialize CTEs (saving them to memory or temp tables), which can impact performance differently than inline subqueries depending on the optimizer.

### Polished answer:

A Common Table Expression (CTE) is an ANSI-SQL standard feature that defines a temporary, named result set that exists solely for the duration of a single `SELECT`, `INSERT`, `UPDATE`, or `DELETE` statement:

- **Syntactic & Readability Advantage:** Subqueries must be nested inside the main query, leading to "rightward drift" and unreadable code when nesting reaches multiple levels. CTEs are declared at the very beginning of the query using the `WITH` keyword, creating a clean top-to-bottom procedural flow.
- **Referential Reuse:** A CTE can be referenced multiple times in the same query (e.g., joining a CTE to itself). With a subquery, you would have to duplicate the identical SQL query block, forcing the query parser to evaluate it multiple times.
- **Recursion:** CTEs support the `RECURSIVE` keyword, allowing them to reference themselves. This is essential for traversing graph and hierarchical tree structures (such as manager-employee trees or bill-of-materials structures) without writing procedural cursors.
- **Optimization Boundary:** In older database versions (like PostgreSQL before v12), CTEs acted as optimization barriers, meaning they were always materialized to disk/memory first. Modern optimizers inline CTEs similarly to subqueries, making performance differences negligible in most cases.

### TL;DR:

A CTE (defined via `WITH`) is a temporary named result set that improves query readability and allows

self-referencing for hierarchical recursion. Unlike subqueries, CTEs can be referenced multiple times in a single query without duplicating code.

### Keyword/Key mappings:

- CTE: Common Table Expression
  - Recursive Query: A query that references itself to process hierarchical data
  - Optimization Barrier: An optimizer setting that prevents a query block from being merged and optimized with the main query
- 

## Q28: How do you write Top-N per group queries in SQL?

### Answer (from given info):

To write Top-N per group queries (e.g., finding the top 3 selling products in each category), there are two common approaches:

1. **Window Functions (Preferred):** Use ranking functions like `ROW_NUMBER()`, `RANK()`, or `DENSE_RANK()` combined with `PARTITION BY` to rank rows within each group, and then filter for ranks  $\leq N$  in an outer query.
2. **Correlated Subqueries:** Compare values within the same group using a subquery. This is used in older database engines that lack window function support but is slower.  
Modern databases prefer window functions because they are cleaner, faster, and easier to maintain.

### Polished answer:

The modern, industry-standard approach to solve Top-N per group queries is using SQL Window Functions with the `PARTITION BY` and `OVER` clauses.

- **Example Implementation (Top 3 highest-paid employees per department):**

```
WITH RankedEmployees AS (  
    SELECT  
        employee_id,  
        name,  
        salary,  
        department_id,  
        ROW_NUMBER() OVER (PARTITION BY department_id ORDER BY salary  
        FROM Employees  
    )  
    SELECT employee_id, name, salary, department_id  
    FROM RankedEmployees  
    WHERE rn <= 3;
```

- **Window Function Breakdown:**

- `PARTITION BY department_id` : Divides the rows into logical groups based on the department.
- `ORDER BY salary DESC` : Sorts the rows inside each department group by salary in descending order.
- `ROW_NUMBER()` : Assigns a sequential integer starting at 1 to each row within its partition.

- **Alternative (Correlated Subquery - Legacy):**

If window functions are unavailable, a correlated subquery can count how many rows within the same group have a higher value:

```
SELECT E1.name, E1.salary, E1.department_id
FROM Employees E1
WHERE 3 > (
    SELECT COUNT(*)
    FROM Employees E2
    WHERE E2.department_id = E1.department_id AND E2.salary > E1.salary
);
```

This legacy approach is an  $O(N^2)$  operation and should be avoided in production.

**TL;DR:**

Top-N per group queries are best written using the `ROW_NUMBER()` window function combined with `PARTITION BY` in a CTE, and then filtering the rank in the outer query. This avoids legacy  $O(N^2)$  correlated subqueries.

**Keyword/Key mappings:**

- Window Function: A function that performs calculations across a set of table rows related to the current row
- Partitioning: Grouping rows within a window function
- ROW\_NUMBER: An incremental row rank starting at 1 for each partition

---

## Q29: What causes duplicate rows after a JOIN, and how do you fix it?

**Answer (from given info):**

Duplicate rows after a JOIN usually occur when a row in one table matches multiple rows in another table. This is common in One-to-Many relationships (such as Customers and Orders). While SQL is behaving correctly based on relational algebra, the result may look incorrect if you are expecting a 1:1 match.

- **Solutions:**

- Ensure you are joining on the correct primary/foreign keys.

- Aggregate the data on the "Many" side using `GROUP BY` before joining.
- Use `DISTINCT` to filter out duplicates, though this can sometimes mask underlying data mapping errors.

### Polished answer:

Duplicate rows in join results are almost always caused by a mismatch in relationship cardinality expectations—specifically, performing a join on a column that contains duplicate values in the joined table:

1. **The Root Cause:** In a One-to-Many relationship (e.g., `Users` join `Orders`), joining Table A with Table B will output a row for every match in Table B. If User A has 5 orders, joining `Users` to `Orders` on `user_id` will duplicate the User A columns 5 times in the result set.

### 2. How to Fix It:

- **Pre-Aggregation (Subquery/CTE):** If you only need aggregated metrics (e.g., total order amount), aggregate Table B *before* joining:

```
SELECT U.user_id, U.username, O.total_spent
FROM Users U
LEFT JOIN (
    SELECT user_id, SUM(amount) as total_spent
    FROM Orders
    GROUP BY user_id
) O ON U.user_id = O.user_id;
```

- **Correct Join Keys:** Ensure that the join condition targets a Unique or Primary Key in the right table if a 1:1 match is expected.
- **Avoid Blind DISTINCT:** Using `SELECT DISTINCT` forces the engine to perform a full-sort deduplication of the entire result set in memory, which is a major performance bottleneck and often masks structural bugs in the join logic.

### TL;DR:

Duplicates are caused by joining on non-unique columns (e.g., 1:N cardinality). Fix this by pre-aggregating the "Many" table using a CTE/subquery before joining, or by refining the join predicate to target unique keys.

### Keyword/Key mappings:

- Cardinality: The numerical relationship between tables (e.g., 1:1, 1:N, M:N)
- Pre-Aggregation: Aggregating data in a subquery before performing a join
- `SELECT DISTINCT`: Deduplicates rows but adds sorting overhead

---

## Q30: What are aggregate functions in SQL? Name a few.

### Answer (from given info):

Aggregate functions in SQL operate on a set of values (or a group of rows) and return a single calculated result. They are commonly used in conjunction with the `GROUP BY` clause.

Common examples include:

- `COUNT ()` : Returns the number of rows or non-NULL values in a column.
- `SUM ()` : Returns the total sum of a numeric column.
- `AVG ()` : Returns the mathematical average of a numeric column.
- `MAX ()` : Returns the maximum value in a column.
- `MIN ()` : Returns the minimum value in a column.

### Polished answer:

Aggregate functions perform mathematical computations on a collection of values across multiple rows to output a single scalar value. Under the hood, the database engine processes these functions by executing a stream aggregation or hash aggregation algorithm:

1. `COUNT (column)` vs `COUNT (*)` :
  - `COUNT (column)` evaluates and counts only the rows where the specified column is **not NULL**.
  - `COUNT (*)` counts the total number of rows in the table or group, including rows with NULL values, because it evaluates the entire row existence.
2. `SUM ()` and `AVG ()` : Perform calculations on numeric columns. They ignore `NULL` values during calculation (e.g., `AVG` divides by the count of non-null rows).
3. `MAX ()` and `MIN ()` : Scan columns (including strings and dates) to find the extreme values. They utilize indexes if available, allowing the optimizer to retrieve the value in  $O(1)$  time without reading the table.

When combined with `GROUP BY`, the engine partitions the rows into buckets and applies the aggregation function independently to each bucket.

### TL;DR:

Aggregate functions (COUNT, SUM, AVG, MAX, MIN) condense multi-row datasets into single summary values. They ignore NULL values (except `COUNT (*)`) and are often optimized using underlying indexes.

### Keyword/Key mappings:

- Stream Aggregation: Calculating aggregates on pre-sorted data (fast)
- Hash Aggregation: Utilizing a temporary hash table to calculate aggregates (memory-intensive)
- Scalar Value: A single numeric or string value

---

## Q31: What is a database cursor and how is it used?

### Answer (from given info):

A cursor in DBMS is a pointer to the result set of a query. While SQL is designed to work in a set-based manner (processing multiple rows at once), a cursor allows for row-by-row processing of query results.

- **Types of Cursors:**

- **Implicit Cursors:** Automatically created by the DBMS for standard SELECT, INSERT, UPDATE, and DELETE operations.
- **Explicit Cursors:** Manually defined, opened, fetched row-by-row, and closed by the database developer.

- **Performance Warning:** Cursors are generally much slower than set-based SQL queries because they require sequential processing, which adds execution and lock overhead. Set-based operations are almost always preferred.

### Polished answer:

A database cursor is a control structure that enables sequential, row-by-row traversal of a query result set. It acts as an iterator or pointer in procedural database programming:

- **How it Works (Explicit Cursor Lifecycle):**

1. **DECLARE** : Defines the cursor's SQL query.
2. **OPEN** : Executes the query and allocates memory (temp tables/buffers) for the result set.
3. **FETCH** : Retrieves the current row from the cursor and assigns it to local variables, advancing the pointer to the next row.
4. **CLOSE** : Releases the cursor's active memory.
5. **DEALLOCATE** : Removes the cursor definition.

- **SDE Performance Context:**

- **The "Row-by-Row" Bottleneck:** Cursors violate the relational set-based design of SQL database engines. They force the engine to run procedural loops, causing excessive context switching between the SQL compiler and the procedural execution engine.
- **Locking Overhead:** Open cursors can hold locks on tables for extended durations during long-running loops, causing transaction log growth and blocking other concurrent transactions.

### TL;DR:

A cursor is a procedural database pointer that processes query rows sequentially (one-by-one).

Because they bypass set-based optimizations and cause context-switching overhead, cursors are slow and should be avoided in favor of set-based SQL.

### Keyword/Key mappings:

- Set-based SQL: Querying entire collections of data at once (highly optimized)
- Procedural DBMS: Programming blocks inside databases (e.g., PL/SQL, T-SQL)
- Context Switching: The CPU overhead of switching execution modes between declarative SQL and procedural loops

---

## Q32: What is the difference between a Superkey and a Candidate key?

### Answer (from given info):

- **Superkey:** A set of one or more attributes that can uniquely identify a row in a table. It can contain unnecessary, redundant attributes.
- **Candidate Key:** A minimal Superkey that uniquely identifies a row with zero redundant attributes. If any attribute is removed from a candidate key, it loses its ability to uniquely identify the row. A table can have multiple candidate keys, and one of them is chosen as the primary key.

### Polished answer:

The distinction between Superkeys and Candidate keys is rooted in the concept of **minimality**:

1. **Superkey (Super set of keys):** Mathematically, any set of attributes  $S$  of a relation schema  $R$  is a Superkey if  $S \rightarrow R$  (meaning the values of  $S$  uniquely determine all other attribute values in the table). A Superkey can contain any number of extra attributes. For example, if `student_id` is unique, then `(student_id, name)` and `(student_id, age, address)` are both valid Superkeys.
  2. **Candidate Key (Minimal Superkey):** A Candidate Key  $K$  is a Superkey with the property of minimality. This means that no proper subset of  $K$  is a Superkey. If you remove even one column from  $K$ , it can no longer uniquely identify a row.
- **Summary Relationship:** All Candidate Keys are Superkeys, but not all Superkeys are Candidate Keys. Candidate Keys are the strict, non-redundant candidates from which the Primary Key is selected.

### TL;DR:

A Superkey is any combination of columns that uniquely identifies a row. A Candidate Key is a minimal Superkey containing only the minimum necessary columns to guarantee uniqueness.

### Keyword/Key mappings:

- Minimality: The property of having no redundant attributes
- Relational Schema: The design definition of a database table
- Determinant: An attribute that uniquely determines another attribute's value

---

## Q33: Explain the difference between a Primary Key and a Unique Key.

### Answer (from given info):

- **Primary Key:**

- Uniquely identifies each record in a table.
- Cannot contain `NULL` values.
- A table can have exactly one Primary Key.
- It automatically creates a Clustered Index (in most databases).
- **Unique Key:**
  - Ensures that all values in a column (or column set) are unique across all rows.
  - Can contain `NULL` values (how many depends on the DBMS; some allow only one, others allow multiple).
  - A table can have multiple Unique Keys.
  - It typically creates a Non-Clustered Index.

### Polished answer:

While both enforce uniqueness constraints, they serve different logical and physical roles in a database:

#### 1. Nullability:

- **Primary Key:** Strictly prohibits `NULL` values. It enforces entity integrity—every entity must have a valid, known identifier.
- **Unique Key:** Allows `NULL` values. In standard SQL, a Unique Key column can contain multiple `NULL` values because `NULL` represents an "unknown" state, and mathematically, `NULL != NULL`. (Note: SQL Server allows only a single `NULL` in a unique column, whereas PostgreSQL and MySQL allow multiple).

2. **Quantity Limit:** A table can have only one Primary Key, which serves as the default row identifier. You can define multiple Unique Keys per table to enforce uniqueness on secondary attributes (e.g., `email`, `social_security_number`).

3. **Physical Storage:** By default, database engines (like InnoDB) organize physical storage using the Primary Key as the Clustered Index. Unique keys are implemented as auxiliary, Non-Clustered B-Tree indexes.

### TL;DR:

A Primary Key uniquely identifies rows, cannot be `NULL`, and dictates the physical storage order (limit: 1 per table). A Unique Key prevents duplicate values but permits `NULL`s, serving as a secondary constraint (limit: multiple).

### Keyword/Key mappings:

- Nullability: The property of allowing or prohibiting `NULL` values
- Entity Integrity: The database rule ensuring every entity is uniquely identifiable
- InnoDB: The default relational storage engine for MySQL

## Q34: Explain the concept of a B-Tree and B+ Tree in DBMS.

### Answer (from given info):

- **B-Tree (Balanced Tree):** A self-balancing search tree data structure that maintains sorted data and allows logarithmic search, insertion, and deletion times. It can store keys and their associated actual data records in both internal nodes and leaf nodes.
- **B+ Tree:** An extension of the B-Tree widely used in database indexing. It differs in two major ways:
  1. It stores actual data records (or pointers to data) **only** in the leaf nodes. Internal nodes store only index keys and pointers to child nodes.
  2. All leaf nodes are linked together in a sequential **linked list**, allowing for highly efficient sequential range scans.

#### Polished answer:

B-Trees and B+ Trees are self-balancing search trees optimized for systems that read and write large blocks of memory (disk I/O). They have a high branching factor (fan-out), which keeps the tree depth low (typically 3 or 4 levels) even for millions of records, minimizing disk lookups:

- **B-Tree Internals:** Every node contains both the index key and the actual data row (or row pointer). This means if a query searches for a key stored in an internal node, it can return the data immediately without traversing down to a leaf node. However, this reduces the capacity of internal nodes, requiring more tree levels and disk lookups for other keys.
- **B+ Tree Internals (The Standard for Database Indexing):**
  - **Internal Nodes:** Contain only routing keys and child pointers. Because no actual data is stored here, internal nodes can hold many more keys. This increases the fan-out and keeps the tree flat.
  - **Leaf Nodes:** Contain all actual data records (or pointers to them) in sorted order.
  - **Leaf Node Linked List:** The leaf nodes are connected via a doubly linked list. In a B-Tree, performing a range scan (e.g., `WHERE age BETWEEN 20 AND 30`) requires traversing up and down the tree repeatedly. In a B+ Tree, the engine traverses down to the first leaf node (`age = 20`) and then simply reads sequentially along the linked list until it hits the end of the range, drastically reducing disk I/O.

#### TL;DR:

B-Trees store data in both internal and leaf nodes. B+ Trees store data *only* in leaf nodes and connect them with a linked list, keeping the tree structure flat and enabling highly efficient sequential range queries.

#### Keyword/Key mappings:

- Fan-out: The number of child pointers a tree node can hold (higher is flatter)
- Balanced Tree: A tree where all leaf nodes are at the exact same depth
- Range Scan: Retrieving a continuous range of ordered values from disk

---

### Q35: Explain the concept of data partitioning in DBMS.

## Answer (from given info):

Data partitioning is the process of dividing large datasets into smaller, more manageable physical segments called partitions to improve performance, scalability, and availability. Each partition can be stored and processed separately.

Types of partitioning:

- **Horizontal Partitioning (Sharding):** Divides data by rows. E.g., storing rows for the year 2020 in one partition, and 2021 in another.
- **Vertical Partitioning:** Divides data by columns. E.g., separating frequently accessed columns into one table and less frequently accessed columns into another.
- **Range Partitioning:** Divides data based on ranges of values (e.g., dates or ages).
- **Hash Partitioning:** Uses a hash function on a key column to distribute rows across partitions.

## Polished answer:

Data Partitioning is a physical design technique that divides a large table or index into smaller, independent physical segments, while maintaining a single logical table interface for applications:

### 1. Partitioning Strategies:

- **Range Partitioning:** Rows are mapped to partitions based on whether a key column's value falls within a specified range (e.g., partitioning an transaction table by month).
- **Hash Partitioning:** A hash function is applied to a partition key (e.g., `HASH(user_id) % 10`). This ensures an even distribution of data across a fixed number of partitions, preventing hot-spots.
- **List Partitioning:** Rows are partitioned based on explicit list values (e.g., partitioning customers by country: `US`, `UK`, `DE`).

### 2. Horizontal vs. Vertical Partitioning:

- **Horizontal (Row-based):** Splits rows across different files or servers (sharding).
- **Vertical (Column-based):** Splits columns of a table into separate physical structures. For example, placing a large, rarely accessed binary column (like `profile_picture_blob`) in a separate table, while keeping basic profile fields in the main table. This reduces the size of the main table's data blocks, speeding up standard queries.

### 3. Key Benefits:

- **Partition Pruning:** The query optimizer reads only the partitions that contain the requested data, bypassing the rest entirely. This turns a full table scan into a targeted scan of a tiny fraction of the data.
- **Maintenance Boundaries:** Administrators can backup, rebuild indexes, or drop old data (e.g., dropping a partition containing data from 5 years ago) in milliseconds without affecting other partitions.

## TL;DR:

Data Partitioning divides large tables into smaller physical chunks (by rows or columns) to improve

search performance via "partition pruning" and make database maintenance (backups, index rebuilds) much faster and safer.

### Keyword/Key mappings:

- Partition Pruning: The optimizer ignoring partitions that do not match the query filter
  - Hot-Spotting: An uneven distribution of write traffic that overloads a single server
  - Sharding: Distributing horizontal database partitions across multiple physical servers
- 

## Q36: What is a hashing technique in DBMS and how does it work?

### Answer (from given info):

A hashing technique maps a data key (like a primary key value) to a fixed-size address using a mathematical hash function. It is primarily used for quick data retrieval, particularly in hash indexes or hash tables, providing fast point-query lookups.

- **How it Works:** A hash function takes a key (e.g., `StudentID = 123`) and calculates a hash value. This value determines the physical bucket or slot where the record is stored. During retrieval, the same hash function is applied to the key to find the exact bucket address instantly, bypassing tree traversals.

### Polished answer:

Hashing in a DBMS is a mechanism used for physical data organization and index retrieval, offering constant-time  $O(1)$  complexity for search operations:

1. **The Hash Function:** A deterministic algorithm that takes an input key of arbitrary size and converts it into a fixed-size integer. This integer represents a physical bucket address on disk or in memory.
2. **Dynamic vs. Static Hashing:**
  - **Static Hashing:** The number of data buckets is fixed. As the database grows, buckets can overflow, requiring **overflow chaining** (linked lists of blocks), which degrades search performance back to  $O(N)$ .
  - **Dynamic (Extensible) Hashing:** The database dynamically grows or shrinks the number of buckets, splitting overflowing buckets and reorganizing the index directory as data size changes. This maintains  $O(1)$  performance.
3. **SDE Practical Context (Limitations):** Hash indexes are incredibly fast for equality comparisons (e.g., `WHERE id = 'user_99'`). However, they do not preserve order. Therefore, they are completely useless for range queries (`WHERE age > 30`) or sorting (`ORDER BY`). B+ Trees remain the default choice for general indexing because of this range limitation.

### TL;DR:

Hashing uses a deterministic function to map keys to physical disk bucket addresses, providing  $O(1)$

time complexity for point lookups. Its main drawback is that it cannot support sorting or range queries because it does not preserve data order.

#### Keyword/Key mappings:

- Overflow Chaining: Linking additional data blocks to a full bucket to handle collisions
  - Point Query: A query searching for an exact key match (e.g., WHERE id = 5)
  - Dynamic Hashing: An adaptive hashing technique where bucket size scales with data volume
- 

## Q37: What are the differences between an ER Diagram and a Relational Schema?

#### Answer (from given info):

- **Entity-Relationship Diagram (ERD):** A conceptual, high-level blueprint that models real-world objects as entities, their attributes, and how they relate to each other. It is used during the database design phase to understand domain concepts.
- **Relational Schema:** A logical, detailed schema that defines the actual structure of a relational database. It represents how data is physically and logically organized in tables, specifying column names, data types, primary keys, foreign keys, and integrity constraints.

#### Polished answer:

ER Diagrams and Relational Schemas represent different phases of the software development lifecycle:

##### 1. Abstraction Level:

- **ER Diagram (Conceptual Model):** A high-level, implementation-independent representation. It focuses on understanding the business domain using entities (rectangles), attributes (ovals), and relationships (diamonds). It does not concern itself with specific databases, data types, or physical storage.
- **Relational Schema (Logical Model):** A database-specific blueprint. It translates the abstract ER diagram into concrete mathematical relations (tables). It defines exact table structures, data types (e.g., `VARCHAR(50)` , `INT` ), primary keys, foreign keys, nullability constraints, and index definitions.

##### 2. Key Translations:

- An **Entity** in an ERD becomes a **Table** in a Relational Schema.
- An **Attribute** becomes a **Column**.
- A **Many-to-Many Relationship** in an ERD is converted into an **Associative Table (Junction Table)** with foreign keys referencing both primary tables.

#### TL;DR:

An ER Diagram is an abstract conceptual model that represents real-world entities and relationships for

business design. A Relational Schema is a concrete logical model that defines actual database tables, data types, and key constraints for physical implementation.

### Keyword/Key mappings:

- Conceptual Model: High-level system design (ERD)
  - Logical Model: Detailed database schema (Relational Schema)
  - Associative Mapping: Translating a many-to-many relationship into a table with foreign keys
- 

## Q38: What is the purpose of the GROUP BY clause in SQL? Explain with execution order.

### Answer (from given info):

The **GROUP BY** clause is used in SQL to partition rows that have identical values in specified columns into summary rows (groups). It is typically used with aggregate functions (like **COUNT** , **SUM** , **AVG** , **MAX** , **MIN** ) to perform reporting and analytical calculations on each group.

- **Example:** Groups employees by department and counts the number of employees in each department.

```
SELECT Department, COUNT(*) FROM Employees GROUP BY Department;
```

- **Execution Order:** FROM/JOIN -> WHERE -> GROUP BY -> HAVING -> SELECT -> ORDER BY -> LIMIT .

### Polished answer:

The **GROUP BY** clause is a powerful relational algebra projection that partitions a dataset into distinct, non-overlapping groups of rows based on matching column values:

1. **The Grouping Mechanism:** When the database engine executes **GROUP BY** , it re-sorts or hashes the rows based on the grouping keys. Every row with identical grouping values is collapsed into a single representative group row.
2. **Aggregation Rule:** Once grouped, any column in the **SELECT** list that is *not* included in the **GROUP BY** clause must be wrapped in an aggregate function. This is because there can be multiple values for that column within a single group, and the engine needs a rule (like **SUM** or **MAX** ) to compress them into a single value.
3. **Execution Pipeline Context:**
  - **FROM/JOIN:** Tables are loaded and combined.
  - **WHERE:** Rows are filtered out early.
  - **GROUP BY:** The remaining rows are partitioned into groups.
  - **HAVING:** The groups themselves are filtered (aggregates evaluated here).
  - **SELECT:** Columns are projected, and remaining aggregate values are calculated.
  - **ORDER BY / LIMIT:** The final collapsed rows are sorted and sliced.

## TL;DR:

**GROUP BY** groups rows with matching column values into single summary rows so that aggregate functions (like COUNT, SUM) can be applied to each partition. It is executed after **WHERE** and before **HAVING**.

## Keyword/Key mappings:

- Hash Aggregation: Using a hash table to assign incoming rows to their respective groups
  - Grouping Key: The column(s) specified in the GROUP BY clause
  - Projection: Choosing specific columns to display in the SELECT statement
- 

## Q39: What is a transaction log in DBMS and why is it important?

### Answer (from given info):

A transaction log (or redo/undo log) is a record that keeps track of all data modifications and transactions executed on a database. It contains start, commit, and rollback events, as well as before-and-after values of modified data.

It is important because, in the event of a system failure, crash, or power loss, the log is used during restart to recover the database to its last consistent state by re-doing committed changes and rolling back uncommitted ones.

### Polished answer:

A Transaction Log (also known as the Write-Ahead Log or WAL) is a sequential, append-only physical file on disk that records all changes made to the database before those changes are written to the actual data blocks. It is the primary mechanism for guaranteeing the **Atomicity** and **Durability** of transactions:

1. **Write-Ahead Logging (WAL) Protocol:** When a transaction modifies a row, the change is first written to the transaction log buffer in memory and flushed to the log file on disk. The actual data page in memory (the dirty page) is written to the disk data file later (asynchronously). This is because writing sequentially to a log is extremely fast (minimal disk head movement), while writing to data files involves slow, random disk I/O.
2. **Crash Recovery (ARIES Algorithm):** If the database server crashes, upon reboot, the recovery manager scans the transaction log:
  - **Redo Phase:** Replays all committed transactions found in the log that were not yet written to the physical database files, ensuring **Durability**.
  - **Undo Phase:** Rolls back all transactions that were active (uncommitted) at the time of the crash, ensuring **Atomicity**.

## TL;DR:

A transaction log is an append-only file on disk that records database modifications *before* they are written to data files (WAL protocol). It ensures crash recovery by replaying committed transactions (Redo) and rolling back uncommitted ones (Undo).

### Keyword/Key mappings:

- WAL: Write-Ahead Logging
  - Crash Recovery: Restoring a database to a consistent state after a failure
  - Dirty Page: A data page in memory that has been modified but not yet written to disk
- 

## Q40: What is the role of a Database Administrator (DBA)?

### Answer (from given info):

A Database Administrator (DBA) is responsible for managing, securing, and overseeing the entire database environment.

Key responsibilities include:

- **Database Design:** Structuring the database for optimal performance.
- **Backup and Recovery:** Planning and executing regular backups, and providing rapid recovery solutions during failures.
- **Performance Tuning:** Monitoring and optimizing queries, indexing strategies, and resource utilization.
- **Security Management:** Managing user access, privileges, and enforcing security policies.
- **Data Integrity:** Ensuring consistency through constraints and validation checks.
- **Troubleshooting:** Identifying and resolving system issues.

### Polished answer:

A Database Administrator (DBA) is a specialized systems engineer responsible for the operational health, security, availability, and performance of an organization's database systems:

1. **Disaster Recovery (DR) and Backups:** Designing robust backup strategies (Full, Incremental, Differential) and validating recovery point objectives (RPO) and recovery time objectives (RTO) to ensure zero data loss during disasters.
2. **Performance Monitoring & Tuning:** Analyzing slow-query logs, execution plans, and system bottlenecks (CPU, disk I/O, lock waits). DBAs optimize indexes, rewrite queries, configure connection pools, and adjust memory allocations (e.g., buffer pool sizes).
3. **Security & Identity Management:** Implementing role-based access control (RBAC), auditing database access, securing connection strings, and managing data encryption at rest and in transit.
4. **Capacity Planning and Scaling:** Monitoring disk growth trends, setting up horizontal sharding, setting up read replicas, and planning hardware upgrades or cloud database migrations.

### TL;DR:

A DBA manages the operational lifecycle of database systems, focusing on disaster recovery (backups), performance tuning (indexing and query optimization), security (access controls), and system scaling.

### Keyword/Key mappings:

- RPO / RTO: Recovery Point Objective (allowable data loss time) and Recovery Time Objective (allowable downtime)
  - RBAC: Role-Based Access Control
  - Read Replicas: Duplicate database nodes used to offload read traffic from the primary master node
- 

## Q41: What is an Entity-Relationship Diagram (ERD)? What are its components?

### Answer (from given info):

An Entity-Relationship Diagram (ERD) is a visual, diagrammatic representation of database entities and their relationships. It is used in database design to model data structures before writing actual code.

Key components:

- **Entities:** Real-world objects represented as rectangles (e.g., Student, Course).
- **Attributes:** Properties of an entity represented as ovals (e.g., Student ID, Name, Age).
- **Relationships:** How entities interact with each other, represented as diamonds (e.g., Student "enrolls in" Course).
- **Example:** A Student entity has attributes (ID, Name, Age) and connects to a Course entity via an "Enrolls" relationship which can have its own attributes (EnrollmentDate).

### Polished answer:

An Entity-Relationship Diagram (ERD) is a conceptual data-modeling tool used to design database schemas. It maps out domain concepts visually using standard Chen's notation or Crow's Foot notation:

1. **Entities:** An object or concept that can store data.
  - **Strong Entity:** Can exist independently (e.g., `Customer` ).
  - **Weak Entity:** Cannot exist without a parent entity (e.g., `Order_Item` cannot exist without an `Order` ). Represented by a double rectangle.
2. **Attributes:** Properties that describe an entity.
  - **Key Attribute:** Uniquely identifies an entity (represented with underlined text, translates to a Primary Key).
  - **Composite Attribute:** Can be broken down into sub-parts (e.g., `Address` splits into `Street` , `City` , `Zip` ).
  - **Multi-valued Attribute:** Can hold multiple values (represented by a double oval, e.g., `PhoneNumbers` ).
  - **Derived Attribute:** Calculated from other attributes (represented by a dashed oval, e.g., `Age` calculated from `DateOfBirth` ).
3. **Relationships:** Structural associations between entities. They express **cardinality** (1:1, 1:N, M:N) and **participation constraints** (total participation vs. partial participation).

## TL;DR:

An ERD is a visual schema design tool. It consists of Entities (independent objects), Attributes (fields describing entities, including unique keys, multi-valued, or derived fields), and Relationships (mapping connections and cardinalities between entities).

## Keyword/Key mappings:

- Strong vs. Weak Entity: Independent entities vs. entities dependent on parent existence
  - Cardinality Constraints: The maximum number of relationship instances an entity can participate in
  - Multi-valued Attribute: An attribute that can hold more than one value for a single entity instance
- 

## Q42: What is the use of the "WITH CHECK OPTION" in SQL views?

### Answer (from given info):

The `WITH CHECK OPTION` is a clause used when creating an updatable SQL view. It ensures that any INSERT or UPDATE operation performed through the view must adhere to the conditions defined in the view's `WHERE` clause. If a user tries to insert or update data through the view that violates these conditions, the operation is rejected with an error.

- **Example:**

```
CREATE VIEW ActiveStudents AS SELECT * FROM Students WHERE Status = 'Active' WITH CHECK OPTION;
```

If a user tries to update a student's status to 'Inactive' or insert a new student with status = 'Pending' through this view, the write operation will fail.

### Polished answer:

The `WITH CHECK OPTION` is a database integrity mechanism that prevents users from performing DML operations through a view that would make the modified rows "disappear" from the view's own result set:

1. **The Problem:** Without this option, if you have a view `ActiveStudents` (defined as `WHERE status = 'Active'`), you could run an update through the view: `UPDATE ActiveStudents SET status = 'Inactive' WHERE student_id = 101;`. The write would succeed, but the updated row would instantly vanish from the view because it no longer matches the `WHERE` condition.
2. **The "WITH CHECK OPTION" Solution:** Adding this clause instructs the database engine to validate every INSERT and UPDATE query executed through the view against the view's filtering criteria. If a write operation would cause a row to fail the view's `WHERE` filter, the engine blocks the write and throws a constraint violation error. This guarantees that all data written through a view remains visible through that same view.

## TL;DR:

**WITH CHECK OPTION** is an integrity constraint on views. It prevents write operations (INSERT/UPDATE) through a view that would modify data such that it no longer meets the view's own **WHERE** filter criteria, throwing an error instead of letting the modified rows disappear.

## Keyword/Key mappings:

- Updatable View: A view through which you can perform INSERT, UPDATE, or DELETE queries
  - Constraint Enforcement: Enforcing validation rules during write operations
  - WHERE Clause Integration: Utilizing view filters as write constraints
- 

## Q43: What are the different phases of the DBMS Query Processing Cycle?

### Answer (from given info):

The DBMS query processing cycle consists of several phases that transform a high-level declarative SQL query into executable physical database actions:

1. **Parsing:** Checking the SQL query for syntax and semantic correctness against the database schema catalog.
2. **Translation:** Converting the SQL statement into an internal logical representation, such as a relational algebra expression or initial execution plan.
3. **Optimization:** Analyzing alternative execution paths and choosing the most efficient execution plan based on cost estimates (evaluating indexes, join algorithms, and CPU/IO costs).
4. **Execution:** The query processor executes the optimized physical plan, retrieves the data from storage, and returns the results to the client.

### Polished answer:

The Query Processing Cycle is the journey an SQL query takes through the database engine to retrieve data with maximum efficiency:

1. **Parser & Lexer:** The SQL string is parsed into tokens. The engine checks for **syntax errors** (grammar) and **semantic validity** (verifying if tables and columns actually exist, and if the user has read privileges). It outputs a structured **Query Parse Tree**.
2. **Query Optimizer (The Core Brain):** The parse tree is converted into logical relational algebra. The optimizer generates multiple alternative physical execution plans. It uses a **Cost-Based Optimizer (CBO)** to evaluate factors such as:
  - Index usage (Index Scan vs. Table Scan).
  - Join algorithms (Nested Loop, Hash Join, or Merge Join).
  - Order of execution.It estimates the physical disk I/O and CPU cost using database statistics, choosing the plan with the lowest estimated cost.

3. **Code Generator / Execution Engine:** The chosen physical plan is compiled into machine instructions or execution byte-code. The Execution Engine traverses the plan (often using the Volcano iterator model) to fetch data blocks from the buffer pool or disk, perform joins/sorts, and return the final dataset.

#### TL;DR:

A query goes through: Parsing (syntax and schema validation), Translation (converting to logical relational algebra), Optimization (cost-based selection of the fastest index and join paths), and Execution (fetching and processing data blocks).

#### Keyword/Key mappings:

- **Cost-Based Optimizer (CBO):** Selecting execution paths based on estimated physical costs (I/O, CPU)
  - **Volcano Iterator Model:** Standard row-by-row pull execution model in database engines
  - **Execution Plan:** The step-by-step physical operations chosen to run a query
- 

## Q44: What are the different types of backups in DBMS?

#### Answer (from given info):

There are four major types of backups in a DBMS to prevent data loss:

- **Full Backup:** Copies the entire database, including all tables, indexes, schemas, and structures. It is comprehensive but slow and requires massive storage.
- **Incremental Backup:** Copies only the data that has changed since the last backup (whether it was a full or incremental backup). It is fast and saves space, but restoring data is slow because you must restore the last full backup followed by every incremental backup sequentially.
- **Differential Backup:** Copies all data modifications made since the last *full* backup. It is faster to restore than incremental backups (only require the full backup + the latest differential backup) but takes more space over time.
- **Transaction Log Backup:** Copies only the transaction log (WAL), enabling point-in-time recovery (replaying database state to an exact second).

#### Polished answer:

An SDE must design backup strategies to meet specific business recovery targets (RPO/RT0):

1. **Full Backup (Baseline):** A complete duplicate of the entire database file system. It serves as the foundation for all other backups.
2. **Incremental Backup (Diff of last backup):**
  - **How it Works:** Captures changes since the absolute last backup of *any* type.
  - **Pros/Cons:** Fast to execute and uses minimal storage. However, recovery is complex and slow because if a crash occurs on Friday, you must restore Sunday's Full Backup, plus

Monday's, Tuesday's, Wednesday's, and Thursday's incremental files in exact chronological order. If any file is corrupted, recovery fails.

### 3. **Differential Backup (Diff of last full backup):**

- **How it Works:** Captures changes since the last *Full* backup.
- **Pros/Cons:** Larger storage footprint than incremental backups, but much faster and safer to restore. To recover, you only need Sunday's Full Backup and Thursday's Differential Backup.

### 4. **Transaction Log Backup:** Periodically backing up the transaction log file (e.g., every 5 minutes).

This is crucial because it allows **Point-in-Time Recovery (PITR)**. If a developer accidentally drops a database at 2:05 PM, the DBA can restore the database to its exact state at 2:04:59 PM by replaying the log.

### **TL;DR:**

Full Backup copies everything. Incremental copies changes since the last backup (fast writes, slow restore). Differential copies changes since the last full backup (moderate writes, fast restore). Transaction Log backups enable Point-in-Time Recovery.

### **Keyword/Key mappings:**

- Point-in-Time Recovery (PITR): Restoring a database to an exact historical timestamp
  - Differential Base: The specific Full Backup that a differential backup references
  - Backup Chain: The sequence of backups required to fully reconstruct a database
- 

## **Q45: How does a DBMS handle concurrency control?**

### **Answer (from given info):**

Concurrency control ensures database consistency when multiple transactions execute simultaneously, preventing conflict anomalies. A DBMS uses several techniques:

- **Locking:** Transactions acquire locks (shared or exclusive) on data items.
- **Two-Phase Locking (2PL):** Involves two phases: growing (acquiring locks) and shrinking (releasing locks) to guarantee serializability.
- **Timestamp Ordering:** Assigns a unique timestamp to each transaction and orders operations based on these timestamps.
- **Optimistic Concurrency Control (OCC):** Transactions read and write data without locks, but during the commit phase, the system validates whether conflicts occurred. If a conflict is found, the transaction is rolled back and retried.

### **Polished answer:**

Concurrency control prevents scheduling anomalies like dirty reads or lost updates in multi-user systems:

#### **1. Pessimistic Concurrency Control (Lock-Based):**

- **Two-Phase Locking (2PL):** Guarantees serializability.
  - **Growing Phase:** Transaction can acquire locks but cannot release any.
  - **Shrinking Phase:** Transaction can release locks but cannot acquire new ones.
- **Rigid/Strict 2PL:** Holds all exclusive (write) locks until the transaction commits or aborts, preventing cascading rollbacks.

## 2. Optimistic Concurrency Control (OCC - Lock-free):

- **Philosophy:** Assumes conflicts are rare.
- **Three Phases:**
  1. *Read Phase:* Transaction reads data and writes modifications to private temporary workspace.
  2. *Validation Phase:* DBMS checks if any other transaction modified the same data since the read phase.
  3. *Write Phase:* If no conflict exists, changes are committed; otherwise, the private workspace is discarded, and the transaction restarts. Excellent for high-read, low-write systems.

## 3. Multi-Version Concurrency Control (MVCC):

- Used by modern databases (PostgreSQL, MySQL InnoDB). Instead of locking a row, the database maintains multiple physical versions of a row with timestamps. Readers do not block writers, and writers do not block readers. Readers access a historical snapshot of the data consistent with their start time.

### TL;DR:

DBMS handles concurrency through: Pessimistic locking (2PL ensures serializability; readers/writers block each other), Optimistic control (OCC validates conflicts at commit time without locks), and MVCC (using row versions so readers and writers do not block each other).

### Keyword/Key mappings:

- **Strict 2PL:** Standard 2PL where exclusive locks are held until transaction commit
- **Optimistic Validation:** The phase where transaction conflicts are checked before writing
- **MVCC Snapshot:** A logical copy of data based on a transaction's start timestamp

## Q46: What are the advantages of using foreign keys in DBMS?

### Answer (from given info):

Using foreign keys provides several key benefits:

- **Enforcing Data Integrity:** Ensures that foreign key values match valid, existing primary keys in the parent table, preventing invalid data entry.
- **Referential Integrity:** Prevents orphaned records in the database by enforcing valid relationships between associated tables.

- **Easy Data Maintenance:** Supports cascading actions (like `ON UPDATE CASCADE` or `ON DELETE CASCADE` ), automatically propagating updates and deletions to child tables.
- **Improved Query Efficiency:** Helps the query engine optimize joins, making queries that combine related tables more efficient and meaningful.

### Polished answer:

In relational design, a `FOREIGN KEY` is not merely a documentation marker but an active constraint engine:

1. **Automated Referential Safeguards:** It physically prevents developers or applications from inserting invalid references. If you attempt to insert an order with a `customer_id` that does not exist in the `Customers` table, the engine rejects the transaction.
2. **Declarative Cascade Orchestration:** It eliminates the need to write manual clean-up code. By configuring `ON DELETE CASCADE` , dropping a user account automatically deletes all corresponding session logs, shopping carts, and orders in a single, atomic operation.
3. **Optimizer Hints (Join Optimization):** The query optimizer analyzes foreign key constraints to make assumptions about table relationships. For instance, in an inner join between Table A and Table B on a foreign key, the optimizer knows with 100% certainty that every matching row in Table B corresponds to exactly one row in Table A, allowing it to select faster join algorithms and simplify query plans.

### TL;DR:

Foreign keys prevent invalid data entry and orphaned records by enforcing referential integrity. They simplify maintenance through automated cascades and assist the query optimizer in selecting more efficient join execution paths.

### Keyword/Key mappings:

- Constraint Validation: The engine checking data validity before committing transactions
- Cascade Deletes: Automatically cleaning up child records when a parent record is deleted
- Query Plan Simplification: Optimizer removing redundant join paths based on foreign key guarantees

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## Q47: Mention the issues with traditional file-based systems that make DBMS a better choice.

### Answer (from given info):

Traditional file-based systems have several critical flaws that DBMS resolves:

- **No Indexing:** Leads to tedious and slow search operations, requiring full-page/full-file sequential scanning.

- **Redundancy and Inconsistency:** Files contain duplicate and redundant data; updating a fact in one place leaves other duplicates outdated and inconsistent.
- **Lack of Concurrency Control:** One operation can lock an entire page or file, preventing other operations from running simultaneously.
- **Other Issues:** Poor security, lack of data isolation, lack of atomicity (partial writes during failures cause corruption), and difficulty in enforcing integrity checks.

**Polished answer:**

A traditional File-Based System relies on the operating system to store raw data files, leaving data organization and retrieval logic entirely to application code. This introduces several major system-level problems:

1. **Data Redundancy and Inconsistency (Isolated Islands of Data):** Different applications maintain their own independent files. Over time, identical data (e.g., customer address) is duplicated across files. If the address changes, some files are updated while others are missed, resulting in inconsistent state.
2. **Procedural Data Access:** File systems lack a declarative query language like SQL. Applications must write nested loops and file-stream pointers to search data, resulting in  $O(N)$  lookup times due to the absence of index structures.
3. **No Transactional Atomicity (The Half-Written File Problem):** If a system crashes mid-operation while writing a file, the file is left partially written and corrupted. There is no automated transaction log or rollback mechanism to restore the file to a clean, consistent state.
4. **Coarse Concurrency Control:** File systems do not support fine-grained (row-level) locking. If two processes attempt to write to the same file, they either overwrite each other's changes (lost update) or one process locks the entire file, blocking others and killing performance.

**TL;DR:**

File-based systems suffer from high data redundancy (leading to inconsistencies), slow procedural search (no indexing), lack of transactional atomicity (risk of file corruption on crash), and poor concurrency (no row-level locking).

**Keyword/Key mappings:**

- Lost Update: A concurrency anomaly where one write overwrites another undetected
- Coarse Locking: Locking an entire file or page instead of a specific record
- Metadata Catalog: A centralized repository of schema definitions (missing in file systems)

## **Q48: What is Data Warehousing? How does it differ from a standard database?**

**Answer (from given info):**

Data Warehousing is the process of collecting, extracting, transforming, and loading (ETL) data from

multiple heterogeneous sources and storing it into a single, centralized database. A data warehouse acts as a central repository where historical data flows from transactional systems and is utilized for data analytics, reporting, and supporting business decision-making processes.

**Polished answer:**

A Data Warehouse is a specialized database system optimized for analytical, read-heavy querying of consolidated historical data:

- **OLTP (Online Transaction Processing) vs. OLAP (Online Analytical Processing):**
  - **Standard Database (OLTP):** Optimized for fast, frequent, and highly concurrent read/write operations. It deals with current operational data, uses highly normalized schemas (3NF) to minimize write latency, and supports daily business transactions.
  - **Data Warehouse (OLAP):** Optimized for complex, long-running analytical queries. It stores consolidated, historical data from multiple sources. It utilizes denormalized schemas (like Star or Snowflake schemas) to minimize joins and accelerate aggregation queries.
- **The ETL Process:**
  - **Extract:** Pulls raw data from various transactional databases and external APIs.
  - **Transform:** Cleans, standardizes, deduplicates, and structures the data.
  - **Load:** Writes the processed data into the centralized data warehouse.

**TL;DR:**

Data Warehousing aggregates historical data from various operational sources via ETL into a central database. Unlike transactional databases (OLTP) optimized for quick writes, data warehouses (OLAP) are denormalized and optimized for complex analytical queries.

**Keyword/Key mappings:**

- ETL: Extract, Transform, Load
- OLAP vs. OLTP: Analytical database design vs. Transactional database design
- Star Schema: A denormalized warehouse schema consisting of central fact tables and surrounding dimension tables

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## Q49: Explain the difference between database Intension and Extension.

**Answer (from given info):**

- **Intension (Schema):** The description of the database structure, defined during the database design phase. It includes tables, columns, data types, and constraints, and it remains mostly unchanged over time.
- **Extension (Database Instance / Snapshot):** The actual data or tuples stored in the database at any specific point in time. The extension represents a snapshot of the database, and its value changes constantly as records are created, updated, or deleted.

### Polished answer:

In database theory, intension and extension distinguish between structural definition and runtime data state:

#### 1. Intension (Metadata/Schema):

- **Definition:** The permanent, declarative definition of the database structure (the schema).
- **Characteristics:** Specified using Data Definition Language (DDL) during the initial design phase. It rarely changes because altering it requires complex migrations. It acts as the mathematical definition of the relation's domains and attributes.

#### 2. Extension (Data/Instance):

- **Definition:** The set of actual tuples (rows) populated within those relations at a given moment in time.
- **Characteristics:** Represents a transient snapshot of the database. It is modified continuously via Data Manipulation Language (DML) as users insert, update, or delete records.

- **Analogy:** In object-oriented programming, **Intension** is equivalent to a **Class** definition, while **Extension** is equivalent to the **Instances (Objects)** created in memory at runtime.

### TL;DR:

Database Intension is the static schema definition (like a Class), while Extension is the dynamic, real-time snapshot of the actual data rows stored in the database (like instantiated Objects).

### Keyword/Key mappings:

- Database Schema: The structural design of the database (Intension)
- Database Instance: The real-time data state of the database (Extension)
- Metadata: Data that describes other data (such as schema definitions)

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## Q50: What are stored functions in DBMS? How do they differ from procedures?

### Answer (from given info):

A stored function is a precompiled set of SQL statements stored in the database that accepts input parameters, executes business logic, and must return a single calculated value.

Differences from stored procedures:

- Functions **must** return a value, whereas procedures are not required to return anything.
- Functions can be called directly within SQL queries (e.g., `SELECT GetEmployeeSalary(ID) FROM Employees`), whereas stored procedures must be invoked manually using `CALL` or `EXECUTE`.

- Functions are typically used to compute values, while procedures are used to perform database modifications (transactions).

**Polished answer:**

Stored Functions are compiled procedural modules designed to extend standard SQL with custom calculation logic:

- **Key Constraints on Functions:**

1. **Strict Return Requirement:** A function must return exactly one value using the `RETURNS <type>` and `RETURN <value>` syntax.
2. **No Side Effects (DML Restrictions):** In most relational engines, functions used inside `SELECT` queries are prohibited from performing data modifications (INSERT, UPDATE, DELETE) or altering database state. This prevents data corruption during parallel query execution. Stored procedures have no such restrictions.
3. **No Transaction Control:** Functions cannot begin, commit, or rollback transactions because they execute within the context of a parent SQL query, which must maintain a consistent read view. Stored procedures can fully manage transaction boundaries.

**TL;DR:**

A stored function is a compiled database routine designed to calculate and return a single value. Unlike procedures, functions can be called directly inside standard SQL queries, but they are restricted from executing transactions or modifying database state.

**Keyword/Key mappings:**

- Side Effects: Changes to database state (such as inserting/updating records)
- Deterministic: A function that always produces the identical output for a given input
- Transaction Boundaries: The start and end points of a transaction (COMMIT/ROLLBACK)